



CHULA **ENGINEERING**
Foundation toward Innovation

COMPUTER

Introduction to Data Science

2110531 Data Science and Data Engineering Tools (2026/1)

Prof. Peerapon Vateekul, Ph.D.

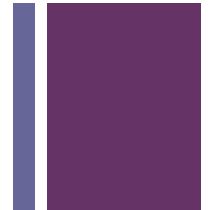
Department of Computer Engineering,
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Peerapon.v@chula.ac.th



Prof. Peerapon Vateekul, Ph.D. (Aj.Pop)

- **Professor**, Department of Computer Engineering, Faculty of Engineering, Chulalongkorn University, Thailand
- **Research Interests:** Applying AI techniques (AI/ML/DL/Gen AI) for real-world applications in **healthcare**, hydrometeorology, transportation, energy, and other domains
- **Member**, NVIDIA AI Technology Center (NVAITC) since 2018
- **Member**, Chula AI – AI Industry Impact Team
- **Senior Researcher**, AI Center of Excellence in Digital and AI for Mental Health (AIMET), Faculty of Engineering, Chulalongkorn University
- Head, **DATA MIND** Lab (Data Analytics, Machine Intelligence, and Knowledge Discovery)
- **Collaborations:** Government agencies (e.g., HII & MEA), industry partners (e.g., TG & SCG), and healthcare institutions (e.g., Chulalongkorn Hospital)





Outline: Understand definitions and terminologies

■ Introduction

- Data is important
- Data Science Definition by Dr.Virote
- Data Science Definition by Aj.Natawut

■ Key Data Science Activities

- Data Science Process
- Types of Data Science Projects
- AI/ML/DL/GenAI
- Data Engineering
- MLOps
- Cloud Technologies

■ Conclusion



Introduction



What is data science? (aka. data analytics)

■ Data

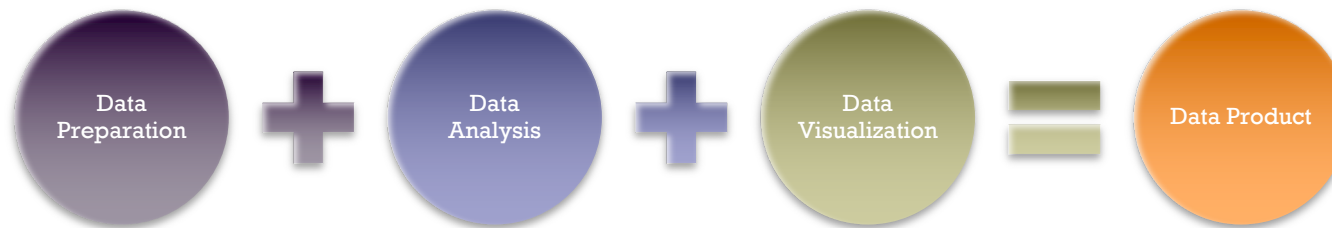
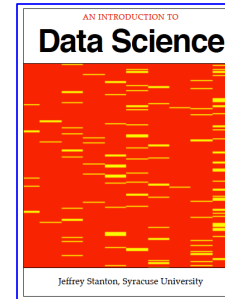
- Facts and statistics collected for reference or analysis

■ Science

- A systematic study through observation and experiment

■ Data Science

- The scientific exploration of data to extract meaning or insight,
- and the construction of software to utilize such insight in a business context.





What is data science? (cont.)

1. Transform data into **valuable insights**
2. Transform data into **data products**
3. Transform data into **interesting stories**



Ta Virot Chiraphadhanakul
Data Scientist, Facebook

Code Mania 2 (01), Jan-2015



1) Transform data into valuable insights

How Obama Raised \$60 Million by Running a Simple Experiment



By Dan Siroker

November 29, 2010

A/B Testing Case Studies, Multivariate Testing, Political Campaign





1) Transform data into valuable insights (cont.)

8

NEW YORK POST

BUSINESS

Amazon introduces next major job killer to face Americans

By James Covert, Linda Massarella and Bruce Golding

December 5, 2015 | 9:59pm | Updated



The Amazon Go storefront

Amazon



Ta Virod Chiraphadhanakul
Data Scientist, Facebook
Code Mania 2, Jan-2015

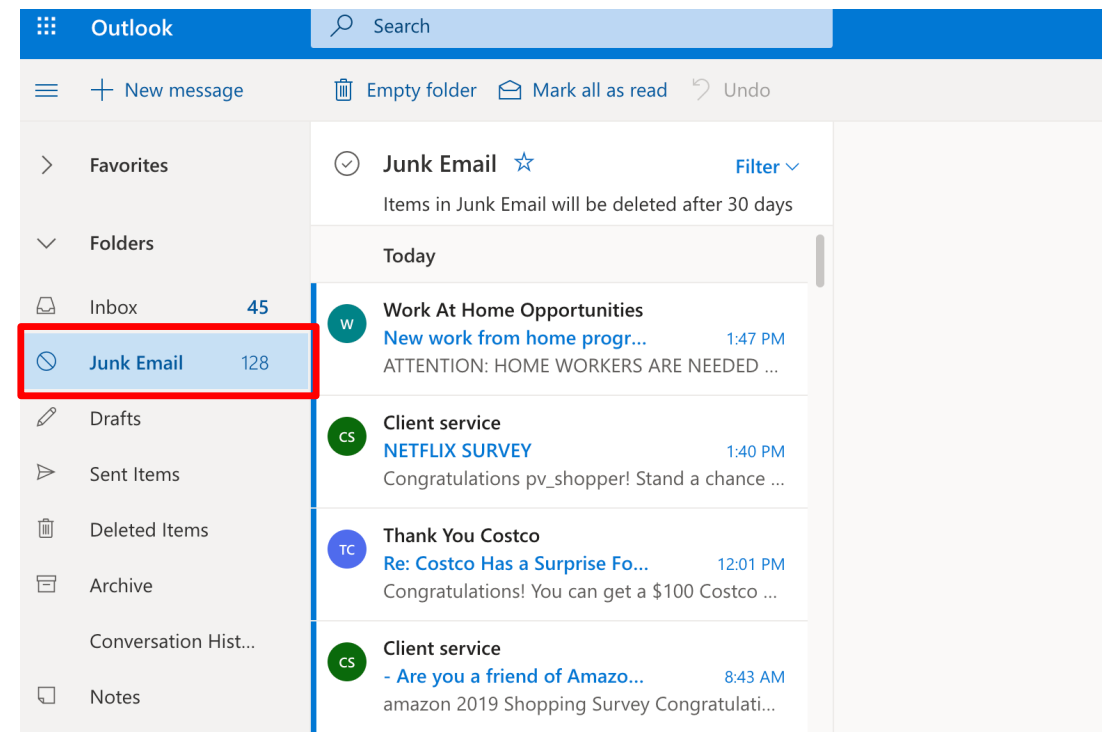
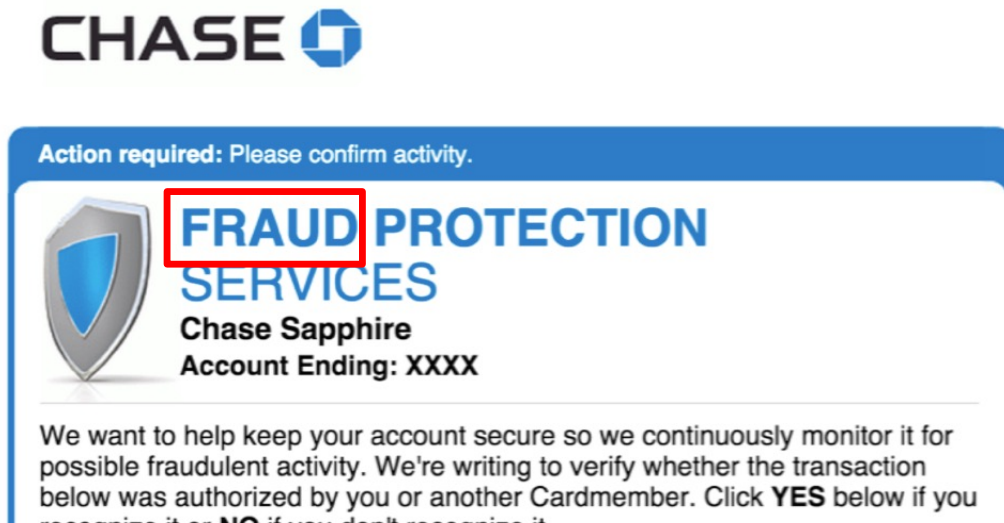
NEW YORK POST POETICS

Amazon's new supermarket will blow your mind — and cost America jobs
Why employment may be optional in the near future
Amazon's futuristic grocery store could spell employment doom
Amazon introduces next major job killer to face Americans
Amazon's latest idea could kill off jobs forever

<http://nypost.com/2016/12/05/amazon-introduces-next-major-job-killer-to-face-americans/>

+ 2) Transform data into data products

9



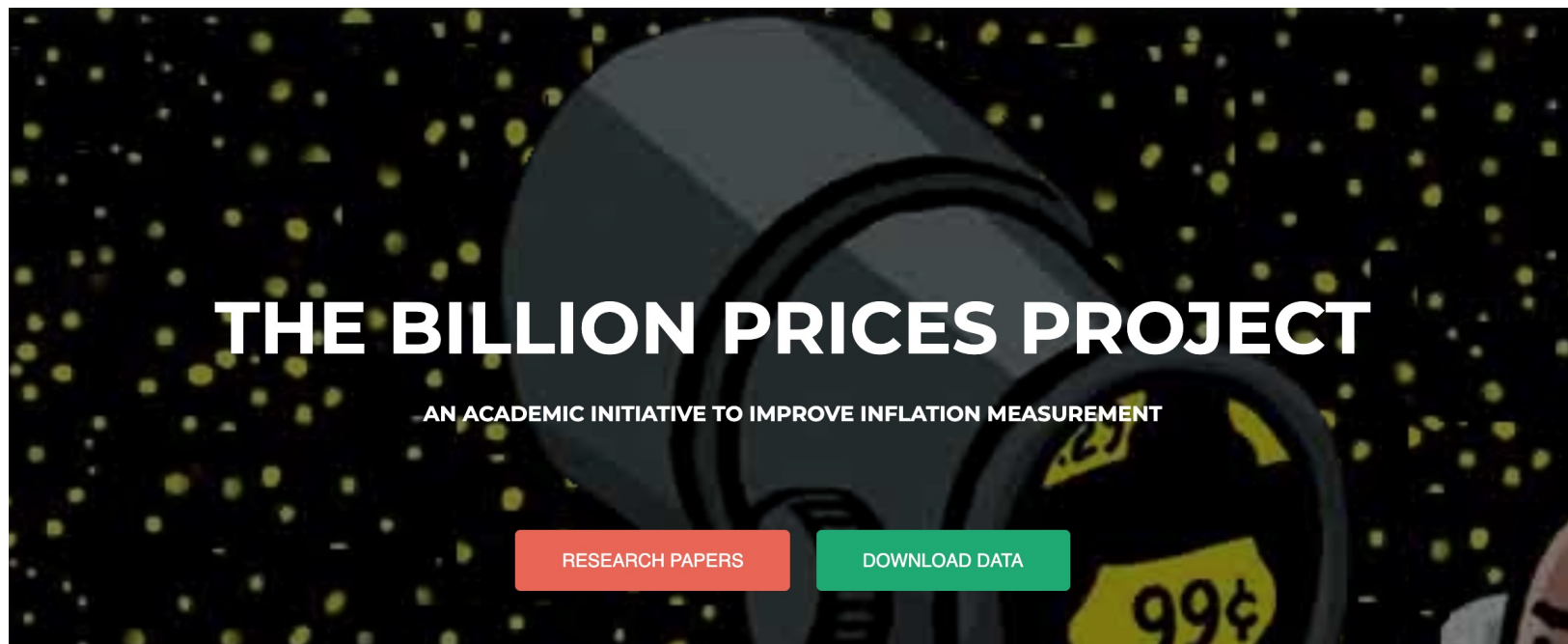


3) Transform data into interesting stories Consumer Price Index (CPI) - Inflation

10

The Billion Prices Project

Home Our Public Data Our Research News



<http://www.thebillionpricesproject.com/>



22-28 Feb. 61

แลกซื้อ สุดคุ้ม

เมื่อซื้อสินค้าที่มีฉลาก
ครบ 79 บาท ขึ้นไป

พร้อมบัตรสมาชิก สามารถแลกซื้อได้ทันทีที่ร้าน
หรือที่ศูนย์บริการลูกค้า (สาขา 1) โทร. 02-010-1000



แลกซื้อ 69 บาท



แลกซื้อ 35 บาท



แลกซื้อ 10 บาท



แลกซื้อ 7 บาท

[illegible]

TESCO Lotus Express

วันที่ 28 ก.ย. - 29 ก.ย. 60

ลดถึงเดือน

อิชิตัน ชาเขียว, ชาผลไม้ ทุกลบ 420 มล.  24 ประหยัดค่า 2 บาท	โค้ก แบบฟอสเฟต, 1.5 ลิตร  19 ประหยัดค่า 1 บาท	ทิฟฟอง แบบฟอสเฟต 300 กรัม  41 ประหยัดค่า 2.5 บาท	โฟร์โมสต์ แบบยูเอที ทุกลบ 225 มล. มี 6 รส  64 ประหยัดค่า 4 บาท
มอคค่า 100% 1 ทุกลบ 110, 14, 20, 22, 100, 120 กรัม  85 ประหยัดค่า 2 บาท	ไอซ์ดี แบบฟอสเฟต 317 กรัม  115 ประหยัดค่า 2 บาท	เนสกาแฟ แบบฟอสเฟต 380 กรัม  198 ประหยัดค่า 15 บาท	ข้าพเจ้าไม่ แบบฟอสเฟต 67 กรัม มี 10 รส  48 ประหยัดค่า 2 บาท
พรมจุ แบบฟอสเฟต 100% 5 ม.  145 ประหยัดค่า 10 บาท	ดิ๊ก แบบฟอสเฟต 5 ม.  149 ประหยัดค่า 10 บาท	ลักส์ แบบฟอสเฟต 500 มล.  89 ประหยัดค่า 5 บาท	ซินธิส แบบฟอสเฟต 500 มล.  96 ประหยัดค่า 5 บาท
โคลเกต แบบฟอสเฟต 150 กรัม มี 2 รส  149 ประหยัดค่า 10 บาท	เคลียร์ แบบฟอสเฟต 300, 320, 340 มล.  105 ประหยัดค่า 5 บาท		
ไลปอน แบบฟอสเฟต 500 มล.  15 ประหยัดค่า 1 บาท	ลอสโอ แบบฟอสเฟต 14, 16, 20 ม.  33 ประหยัดค่า 2 บาท	เอสโก้ แบบฟอสเฟต 800, 900 กรัม  29 ประหยัดค่า 2 บาท	มายไฟ แบบฟอสเฟต 20, 24, 34, 102, 112 ม.  139 ประหยัดค่า 20 บาท
ดาวน์ แบบฟอสเฟต 400, 700 มล.  53 ประหยัดค่า 10 บาท	บรีส แบบฟอสเฟต 900 กรัม  47 ประหยัดค่า 5 บาท	แอกเทค แบบฟอสเฟต 800, 900 กรัม  49 ประหยัดค่า 5 บาท	โอบี แบบฟอสเฟต 1400, 1700 มล.  115 ประหยัดค่า 24 บาท
		ไฮยิเอนท์ แบบฟอสเฟต 600 มล.  13.50 ประหยัดค่า 3.50 บาท	

Trusted Brand
Seal of Excellence

ทิปส์ ได้ รับการยกย่องเป็น Trusted Brand
ซึ่งเป็นสุดยอดแบรนด์ที่พิสูจน์ให้เห็นความเชื่อมั่นจากที่สุด
ต่อเนื่องกันถึง 5 ปี สาขาเปอร์สเบิร์กเกิด จากนิยสาร ฮีโร่ดอร์ ไตรสาร

สินค้าใน Source Unlimited Foodies มีนะ

90 บาท - 12 บาท 250%

Red Hot

19
ข้าวทอดรสไก่บ้าน
1 ห่อ 150 กรัม

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บาท

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1 ห่อ 150 กรัม

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1 ห่อ 150 กรัม

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บาท

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ข้าวทอดรสไก่บ้าน
1 ห่อ 150 กรัม

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บาท

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1 ห่อ 150 กรัม

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บาท

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1 ห่อ 150 กรัม

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บาท

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บาท

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ข้าวทอดรสไก่บ้าน
1 ห่อ 150 กรัม

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บาท

Tops Super

ทิปส์ ได้ รับการยกย่องเป็น Trusted Brand ซึ่งเป็นสุดยอดแบรนด์ที่พิสูจน์ให้เห็นความเชื่อมั่นจากที่สุดต่อเนื่องกันถึง 5 ปี สาขาเปอร์สเบิร์กเกิด จากนิยสาร ฮีโร่ดอร์ ไตรสาร



The Billion Prices Project: Using Online Prices for Measurement and Research *

Alberto Cavallo
MIT and NBER

Roberto Rigobon
MIT and NBER

This Version: April 8, 2016

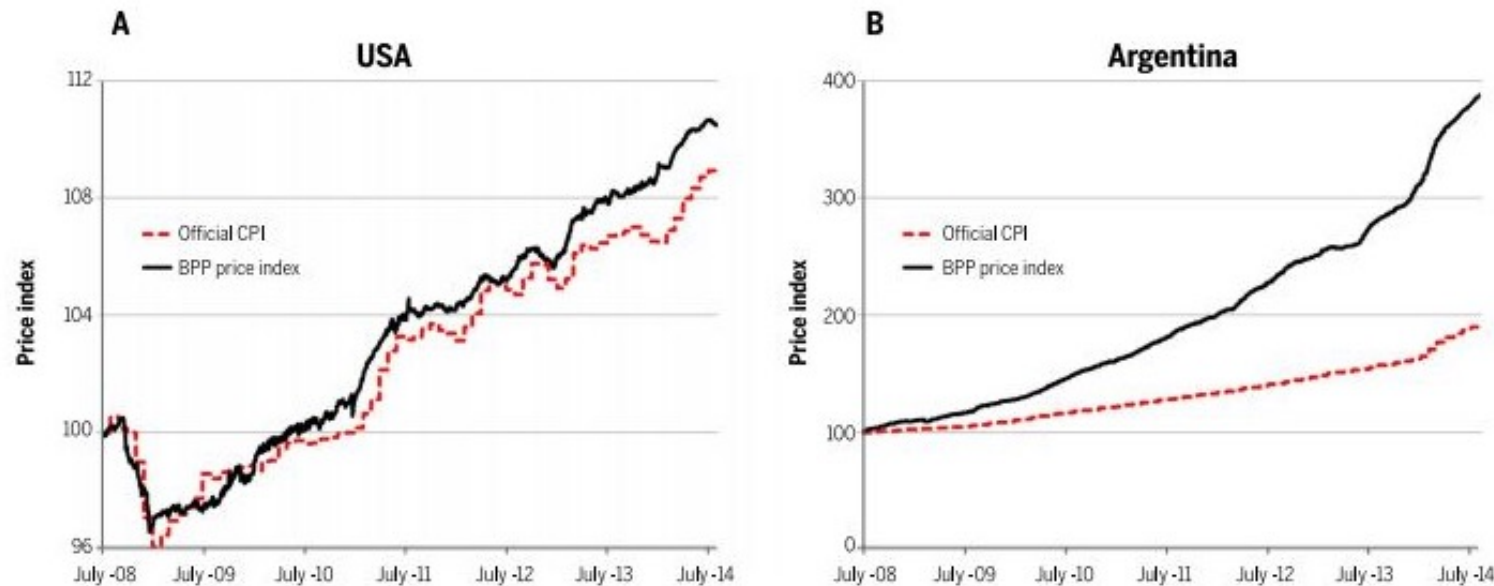
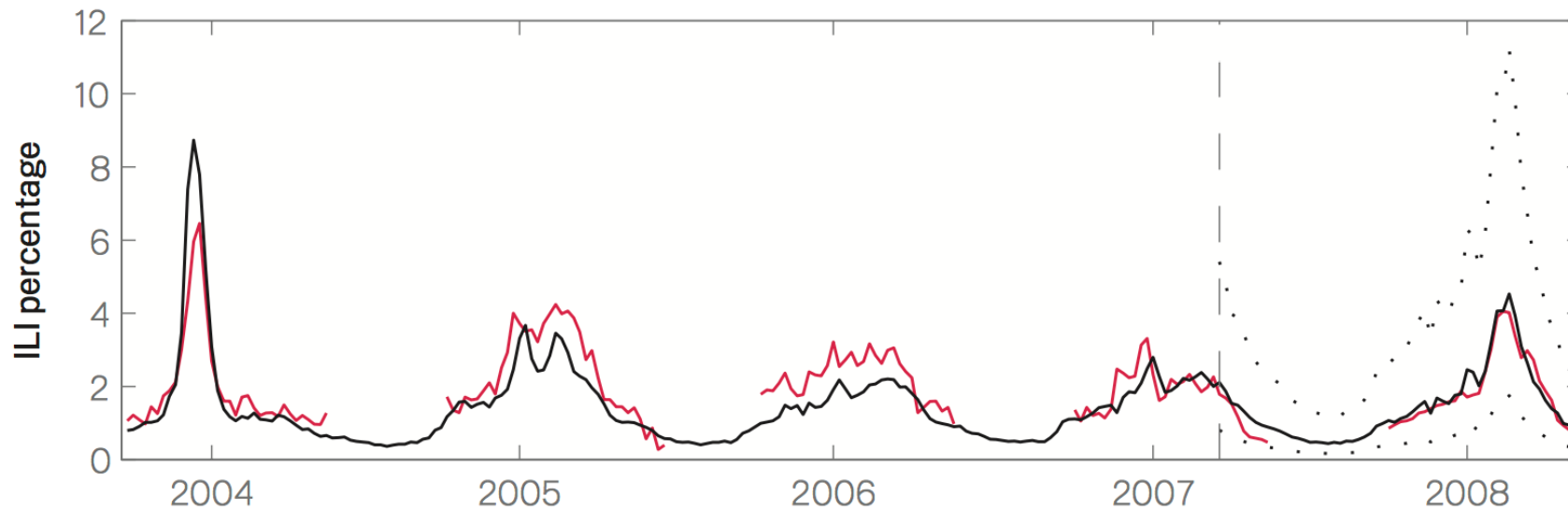


Fig. 2. BPP price index. Dashed red lines show the monthly series for the CPI in the United States (A) and Argentina (B), as published by the formal government statistics agencies. Solid black lines show the daily price index series, the "State Street's PriceStats Series" produced by the BPP, which uses scraped Internet data on thousands of retail items. All indices are normalized to 100 as of July 2008. In the U.S. context, the two series track

each other quite closely, although the BPP index is available in real time and at a more granular level (daily instead of monthly). In the plot for Argentina, the indices diverge considerably, with the BPP index growing at about twice the rate of the official CPI. [Updated version of figure 5 in (18), provided courtesy of Alberto Cavallo and Roberto Rigobon, principal investigators of the BPP]



Google Flu Trend



Ginsberg, Jeremy; Mohebbi, Matthew H.; Patel, Rajan S.; Brammer, Lynnette; Smolinski, Mark S.; Brilliant, Larry (19 February 2009). "Detecting influenza epidemics using search engine query data". *Nature*. **457** (7232): 1012–1014.



What are they using data science for?

1. Measurement
2. Insights
3. Data Products

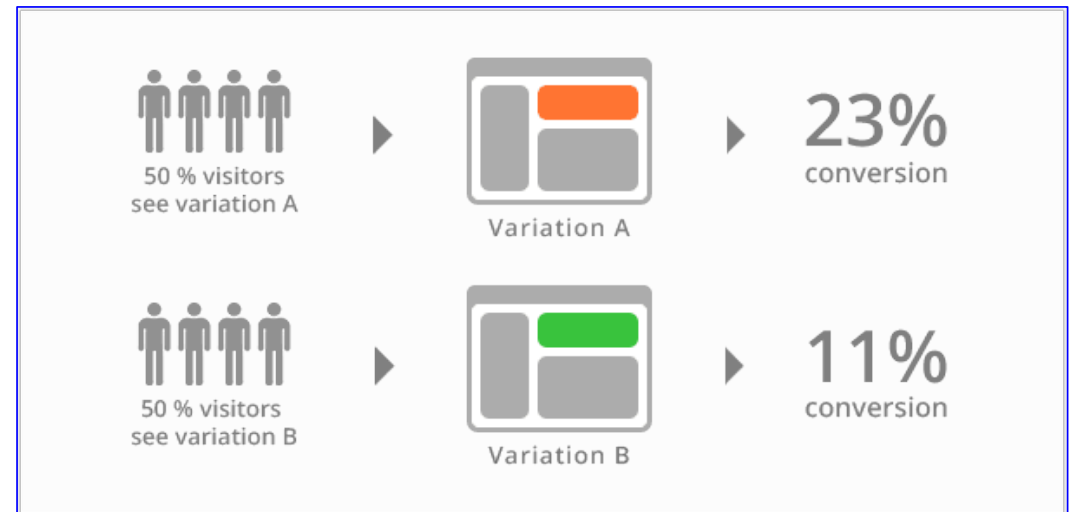




1) Measurement

15

- To **make a decision** based on data (aka. Benchmarking)
- Turning qualitative information into **quantitative values** (metrics or indicators)
- Support comparison:
 - Between options (e.g., which notebook to buy)
 - Before vs. after (e.g., tuning, upgrades)
 - **A/B Testing**



Source: <https://vwo.com/ab-testing/>



Example: SimCity

1. Remove product banner: SimCity sees 43% more conversions without hero banner image

16

Control



Variation



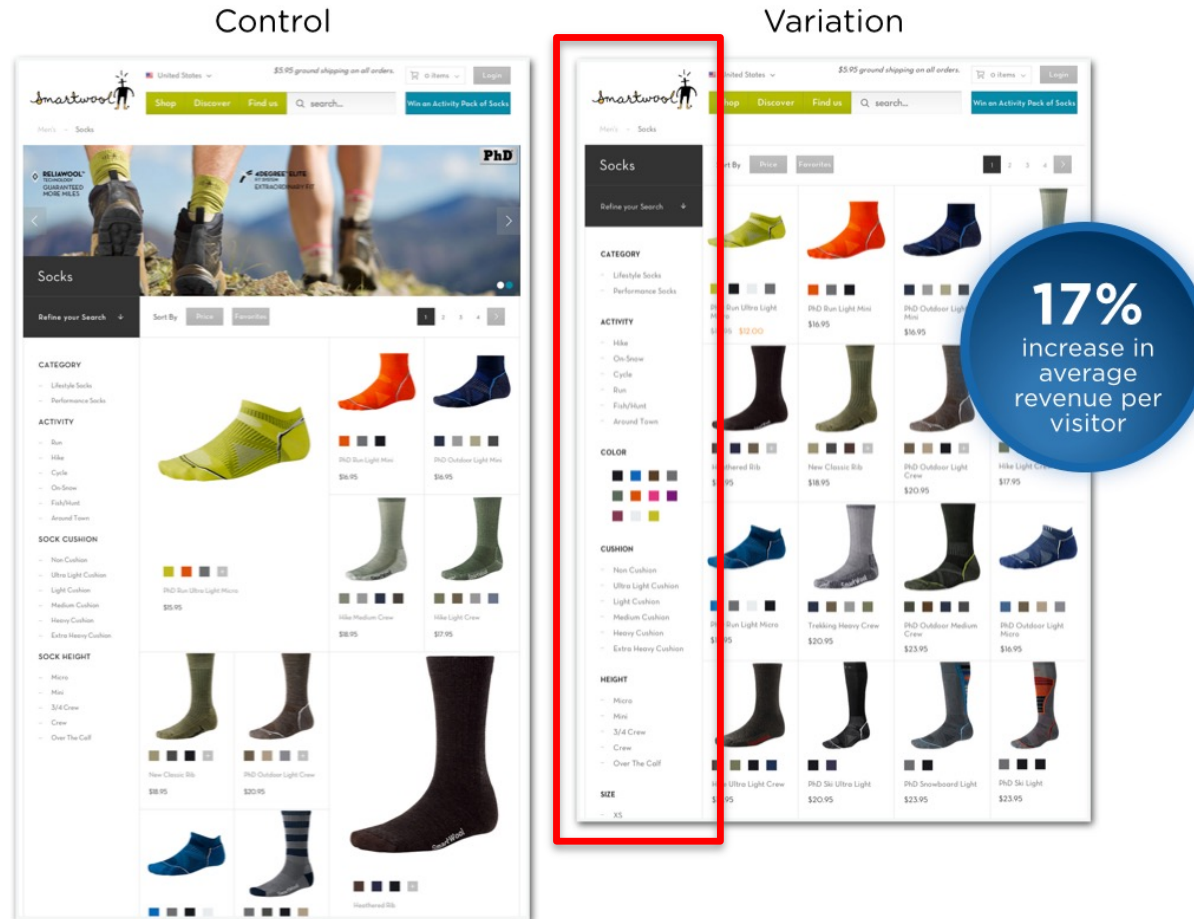
Source: <https://blog.optimizely.com/2015/06/04/ecommerce-conversion-optimization-case-studies/>



Example: SmartWool

3. Use a well-defined grid layout for your online shopping experience: Uniform product page images increase ARPV 17% for SmartWool

17



Source: <https://blog.optimizely.com/2015/06/04/ecommerce-conversion-optimization-case-studies/>



2) Insights

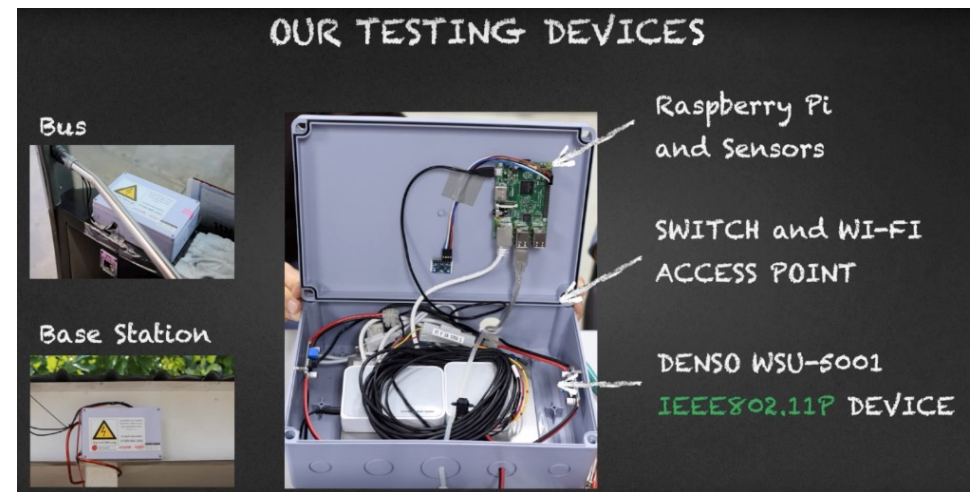
- **Good understanding of user behavior** can lead to new product development or improvements of the existing products
- **Financial startup** -- Typing with proper capitalization indicates creditworthiness
 - Online loan applicants who complete the application form with the correct case are more dependable debtors
- **Starbucks** use customer purchase information from My Starbucks Mobile Apps to figure out new products



Example: Tracing Traffic



19

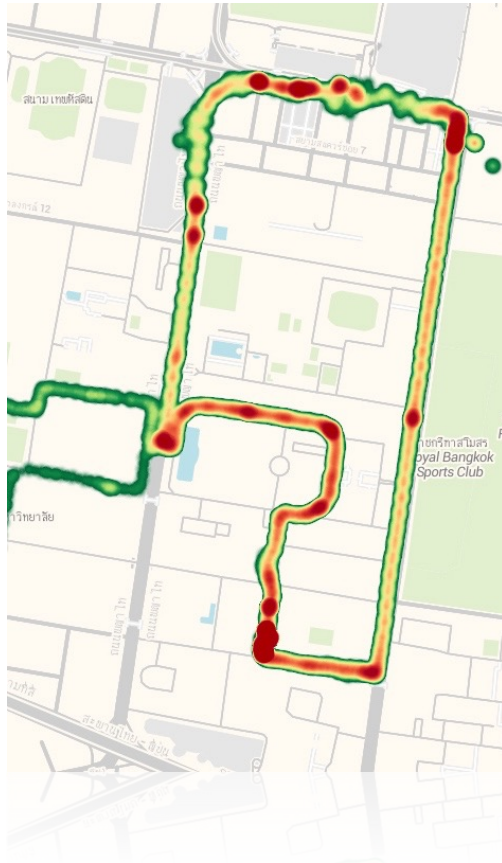




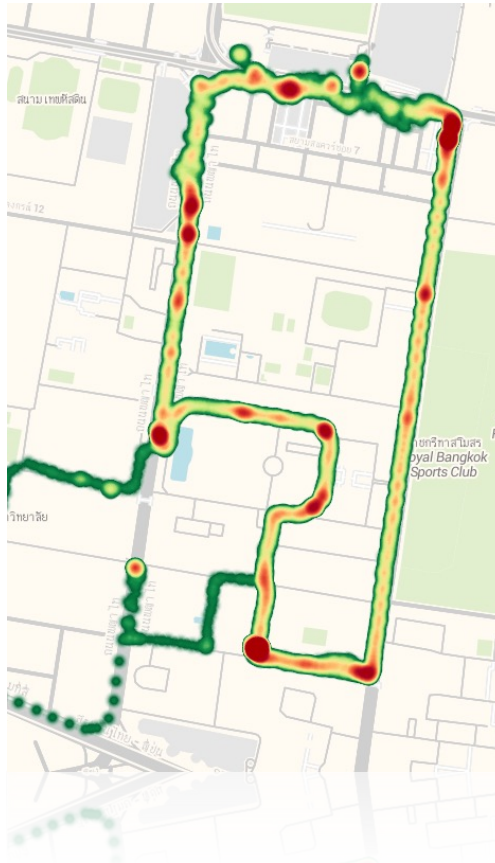
GPS Average Speed

20

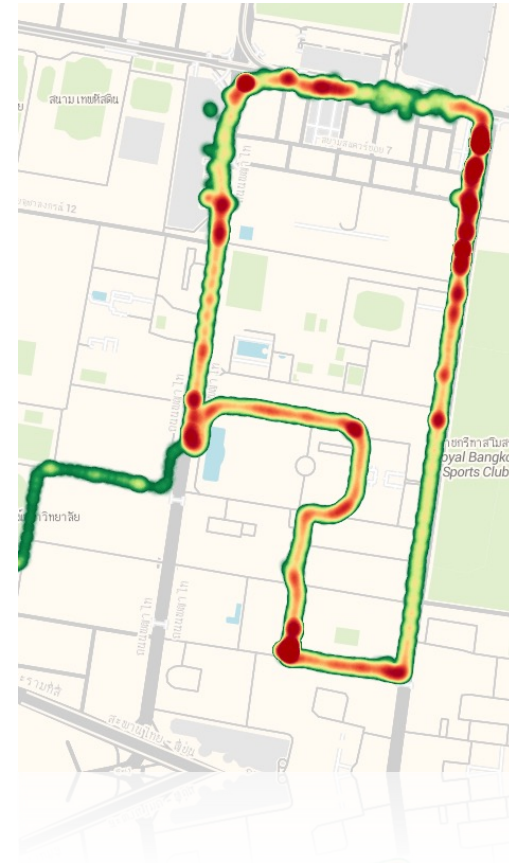
6:00-10:00



10:00-15:00



15:00-18:00

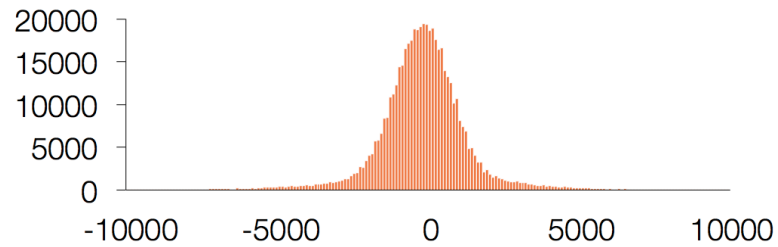




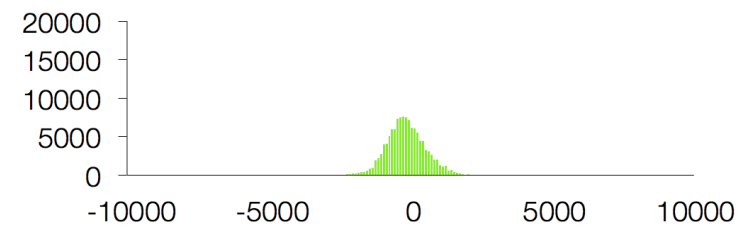
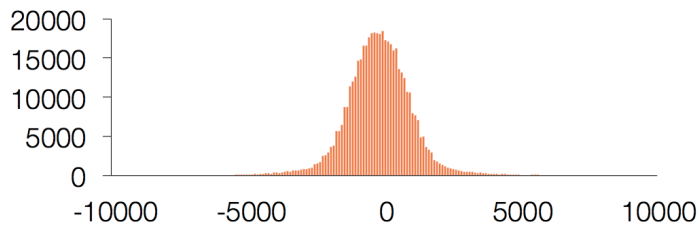
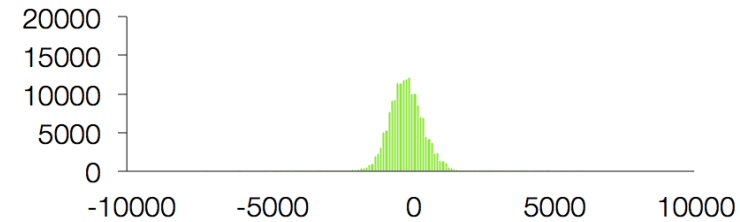
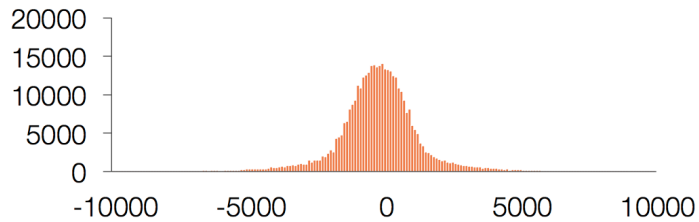
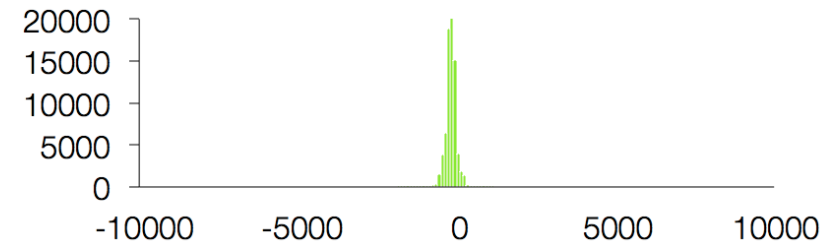
Bus Drivers' Behaviors

21

Bus A



Bus B



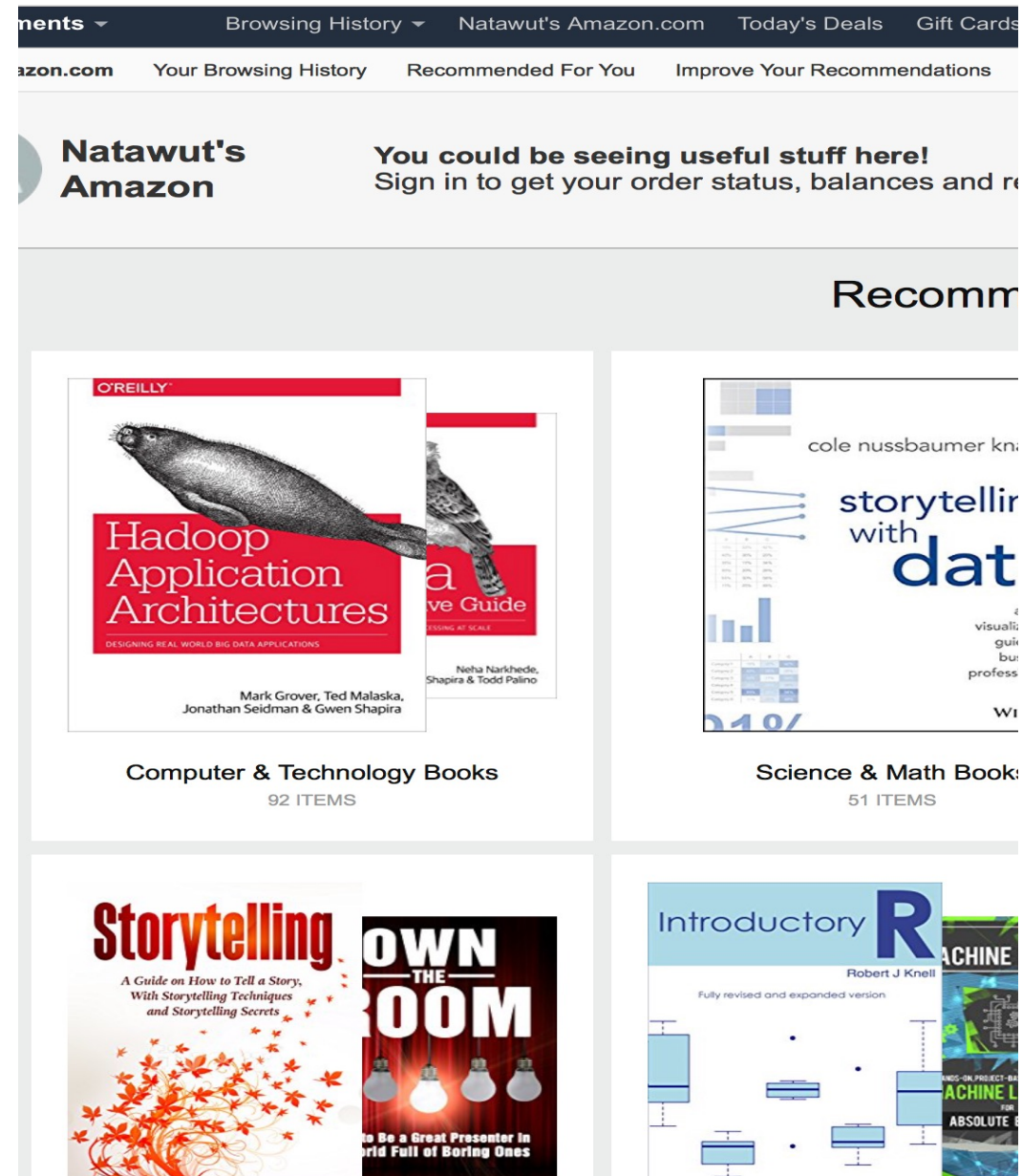


3) Data Products

- An application or system that uses data to provide “**intelligent**” products or services, which create more data that can be further used
- **Machine Learning** plays an important role in building great data products

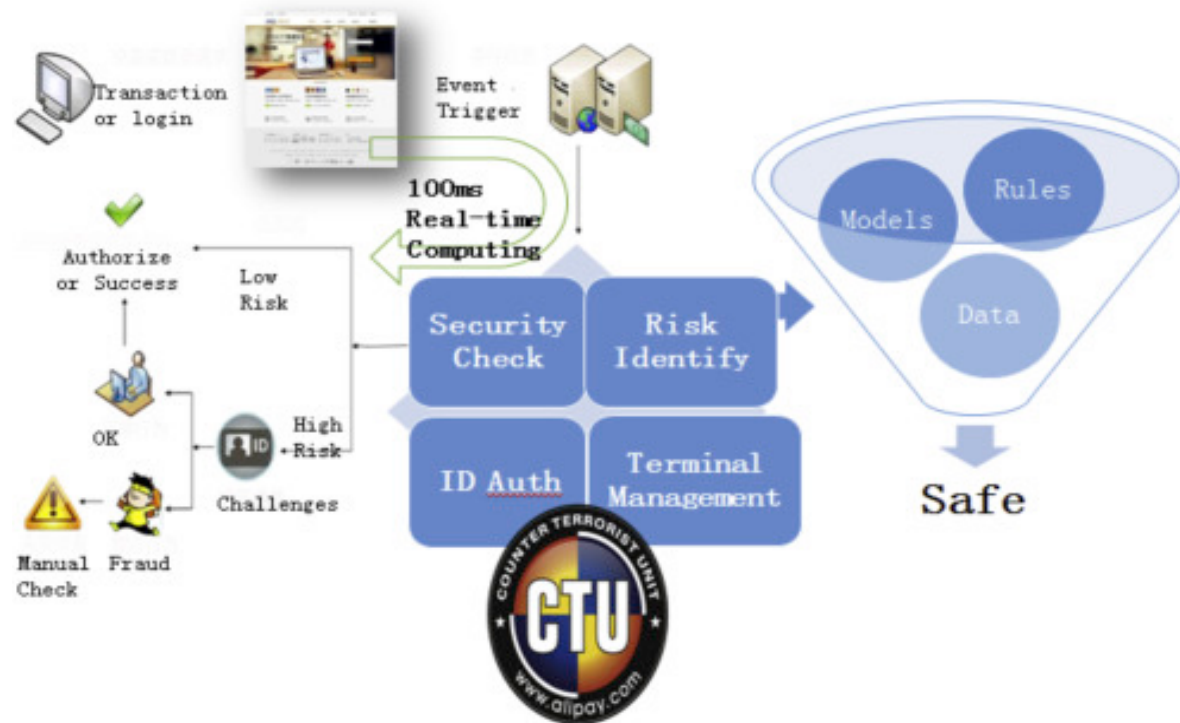
Example: Amazon Recommendation

- Amazon sells 480M products (485k new products per day)
- Use recommendation systems to bring products to customers
- Analyze data from 300M customers
 - Purchase history
 - Reviews / Ratings
 - Search history
 - Views





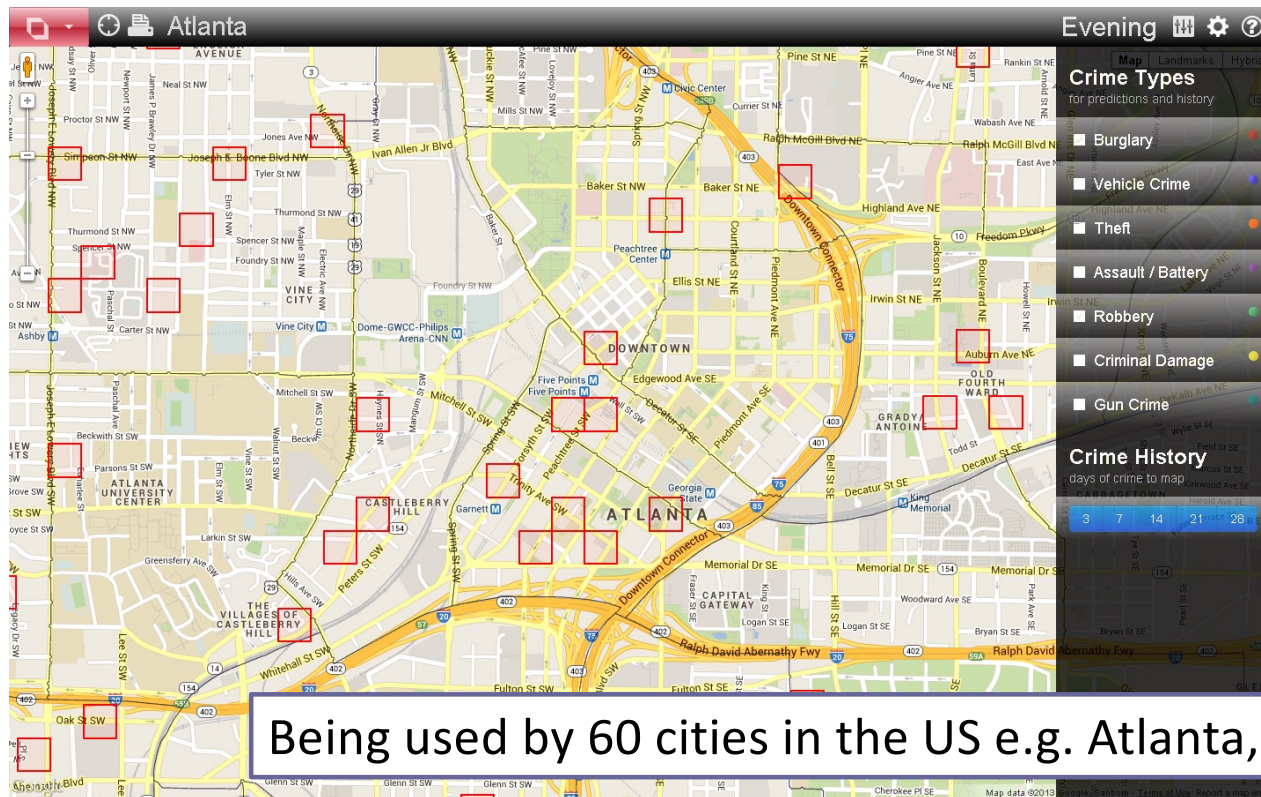
Case study: Alibaba Fraud Detection



Source: <http://www.sciencedirect.com/science/article/pii/S2405918815000021>



Case study: Predictive Policing

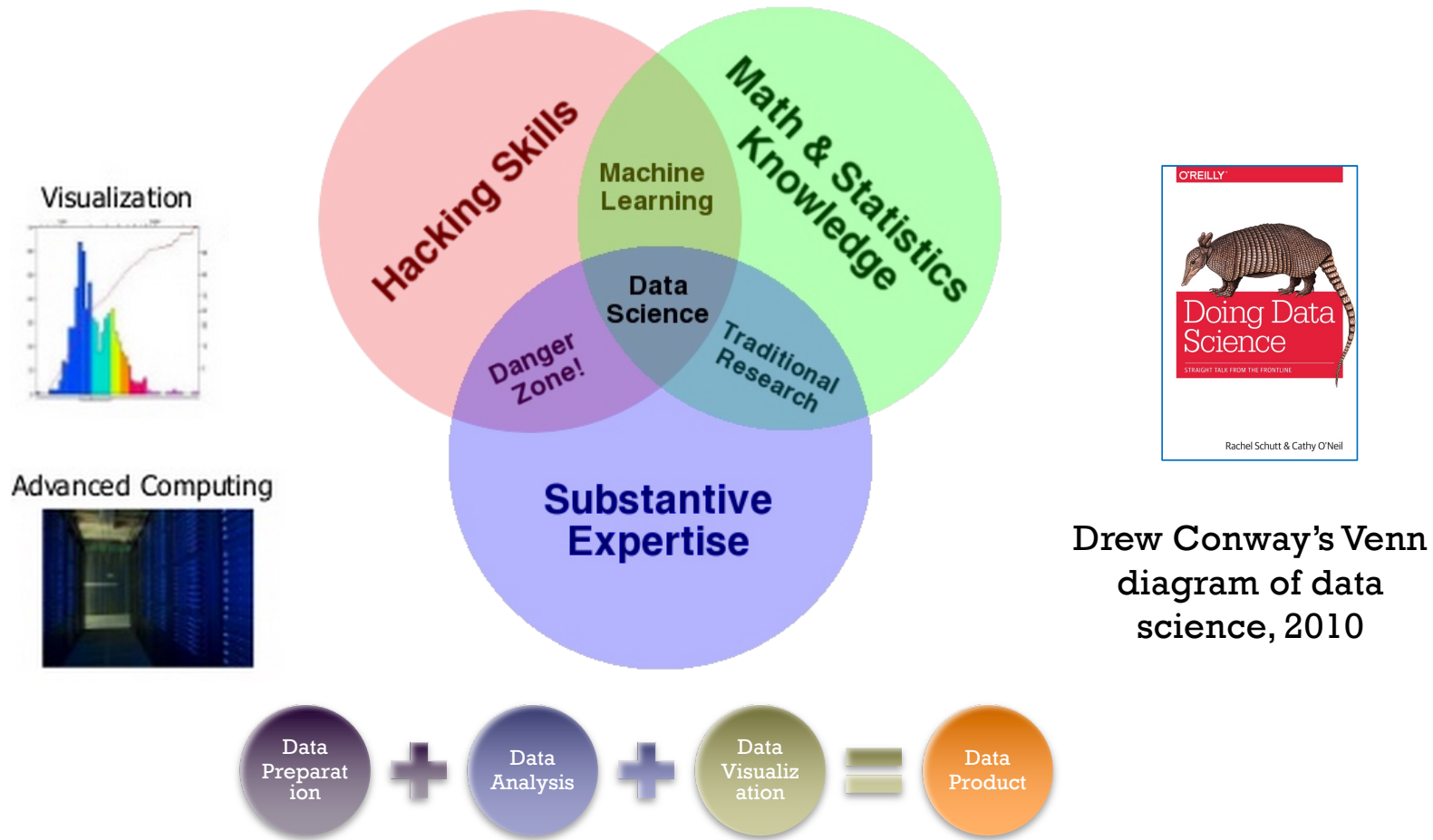


Source: <http://www.forbes.com/sites/ellenhuet/2015/02/11/predpol-predictive-policing>



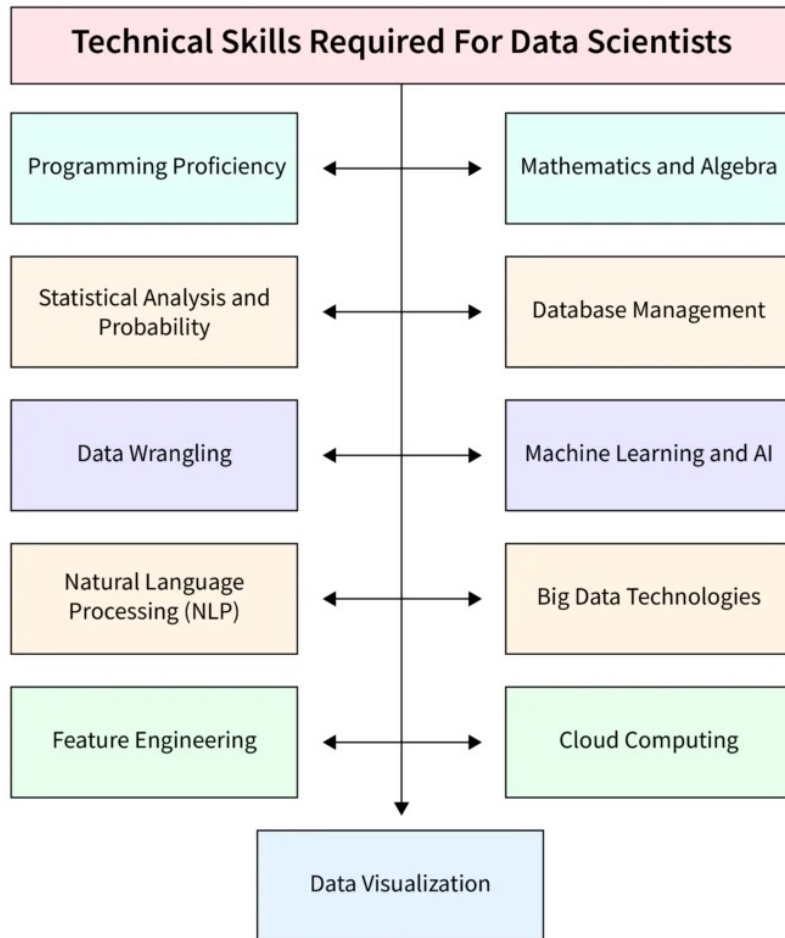
Drew Conway's Data Science Venn diagram (Skills)

26



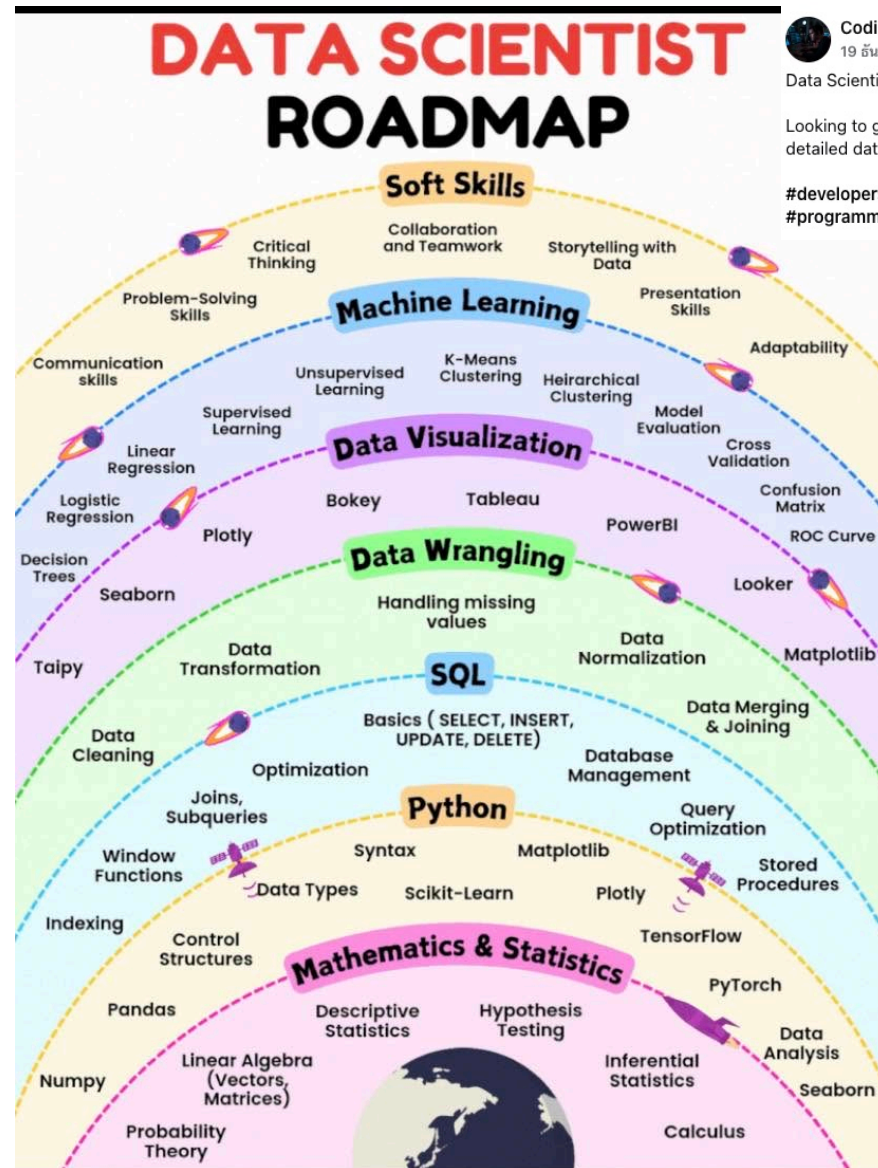
Chula Data Science

Top 20 Data Scientist Skills You Need in 2025



SCALER

Ref Link
Ref Link



Coding Tips
19 ธันวาคม 2025

Data Scientists Roadmap

Looking to get into tech in 2026? This is a detailed data science roadmap

#developers #tips #softwareengineer #programming #datascience

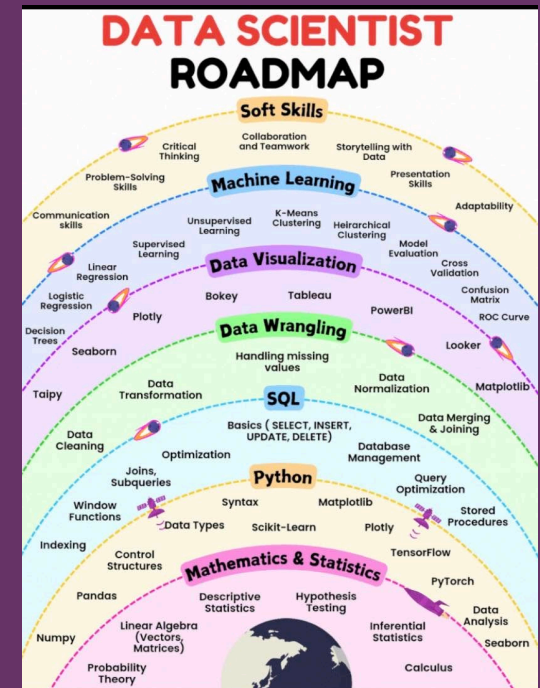


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Key Data Science Activities

Data Science Process
Types of Data Science Projects
AI/ML/DL/GenAI
Data Engineering
MLOps
Cloud Technologies

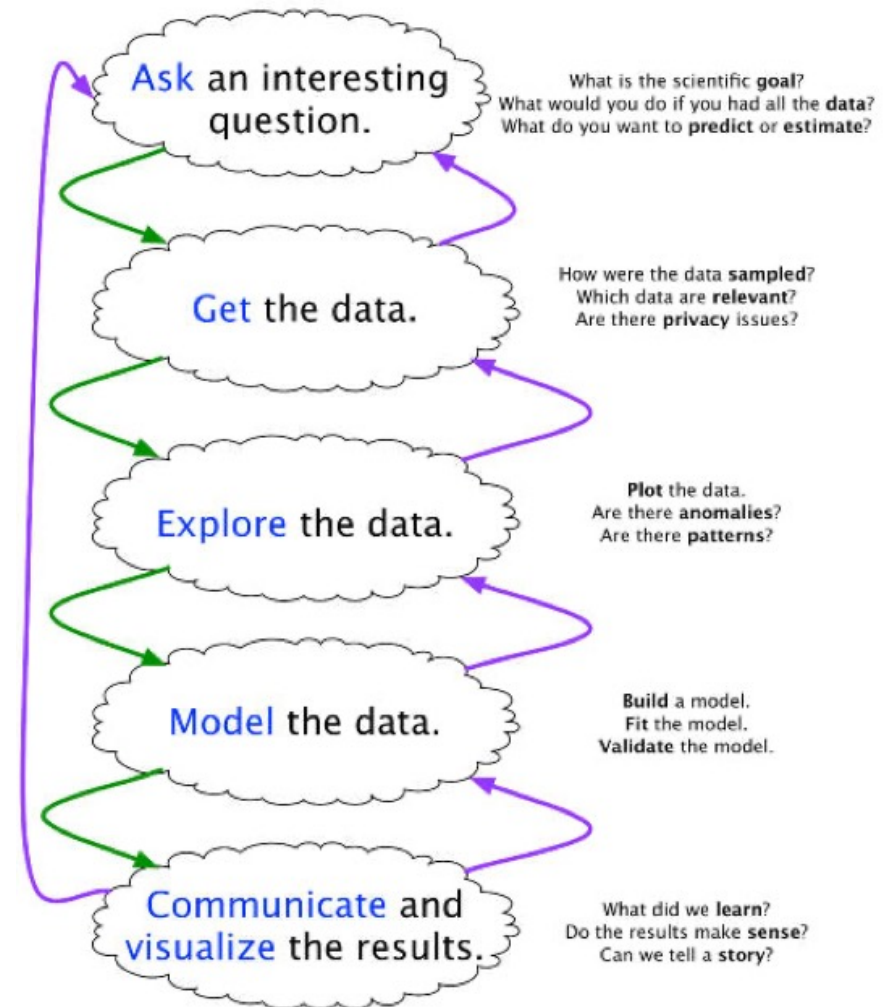




Data Science Process



The Data Science Process



Joe Blitzstein and Hanspeter Pfister, created for the Harvard data science course <http://cs109.org/>.

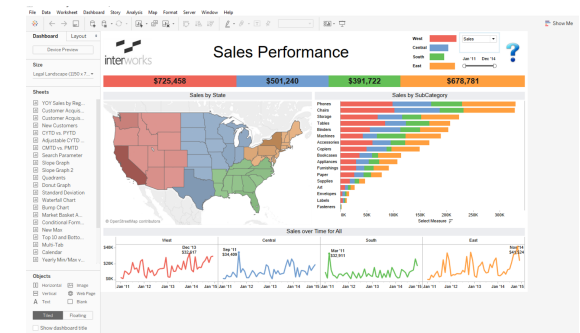
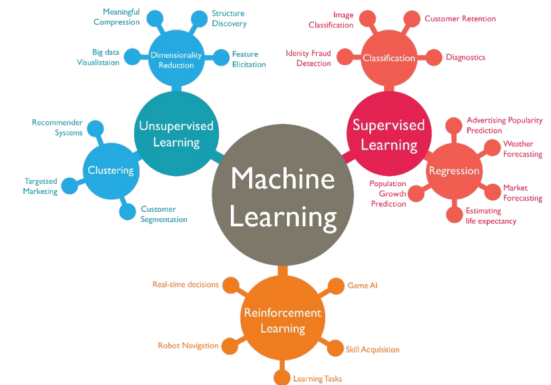
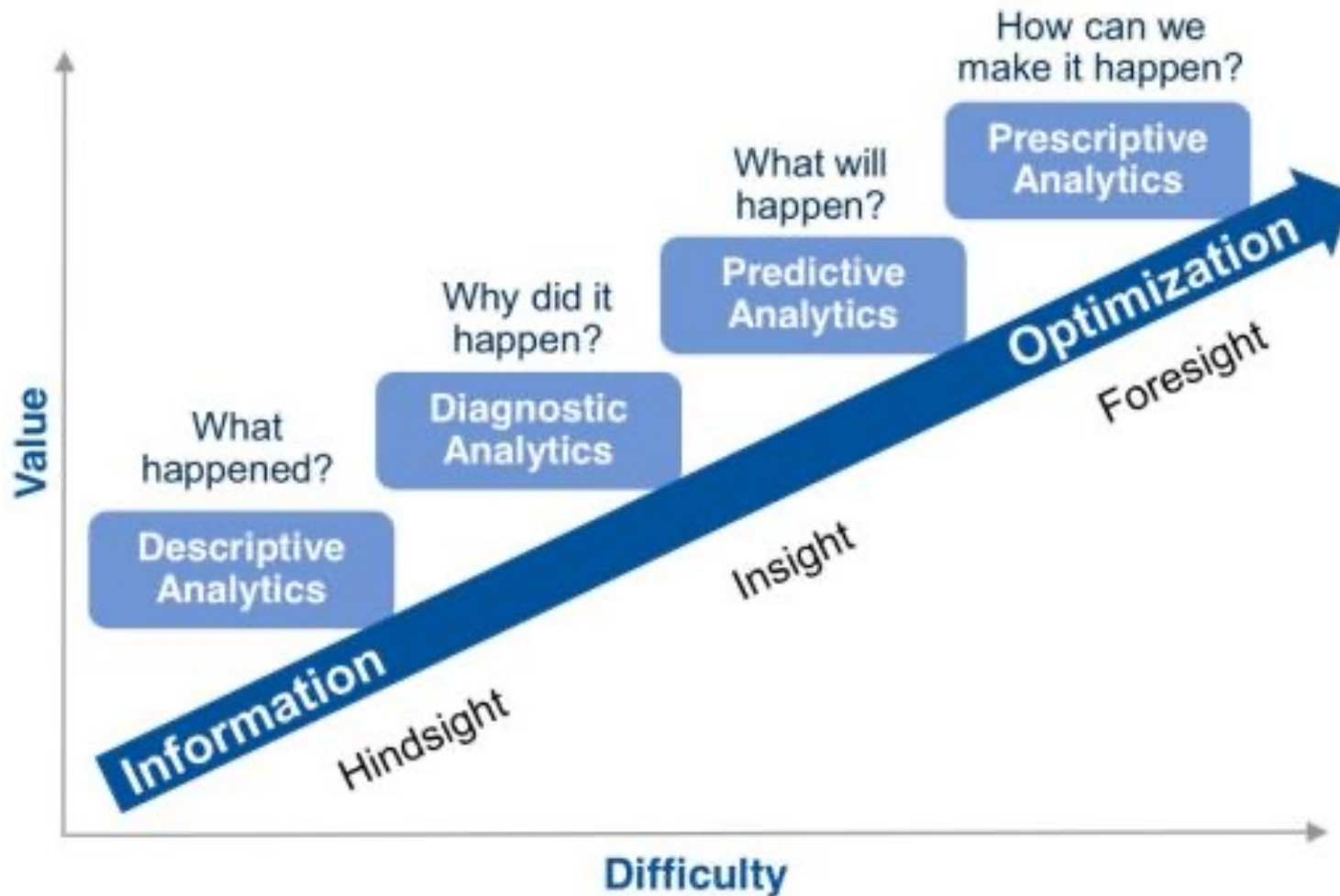
Dr. Virote

1. Transform data into **valuable insights**
2. Transform data into **data products**
3. Transform data into **interesting stories**

Aj. Natawut

1. Measurement (**decision**)
2. Insights (**knowledge**)
3. Data Products (**innovation, intelligent**)

Data Analytics (Data Science)





Types of Data Science Projects

31

Valuable insights

- Data visualization
- Analytical skills & storytelling
 - Infographic



Advanced analytics

- AI/Machine Learning/Deep Learning
 - Prediction, Forecasting, Clustering, etc.



Ask a question about your data

Total Stores
NEW & EXISTING STORES

104

This Year's Sales
NEW & EXISTING STORES

\$22M

This Year's Sales
BY CHAIN



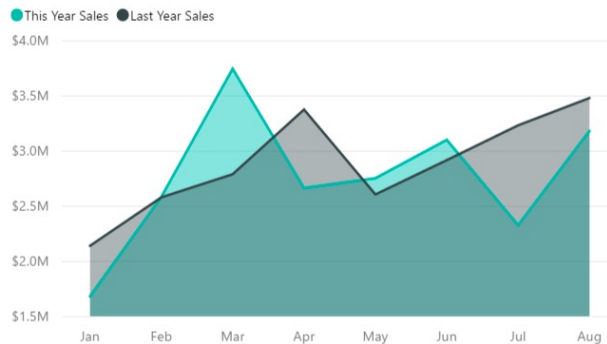
New Stores, New Stores Target
YEAR TO DATE



This Year's Sales
NEW STORES ONLY

\$2M

This Year's Sales, Last Year's Sales
BY FISCAL MONTH



Total Sales Variance %, Sales Per Sq Ft, This Year's Sales
BY DISTRICT



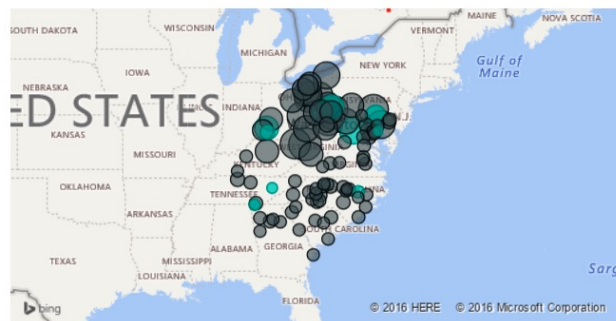
New Stores
NEW STORES ONLY

10

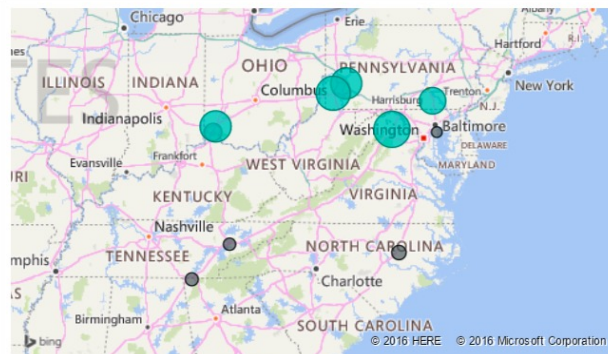
Stores Opened This Year
BY OPEN MONTH, CHAIN



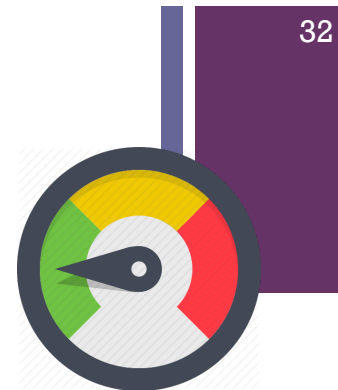
This Year's Sales
BY POSTAL CODE, STORE TYPE



This Year's Sales
BY CITY, CHAIN



Sales Per Sq Ft
BY NAME



The 2026 Gartner® Magic Quadrant™ for Analytics & Business Intelligence



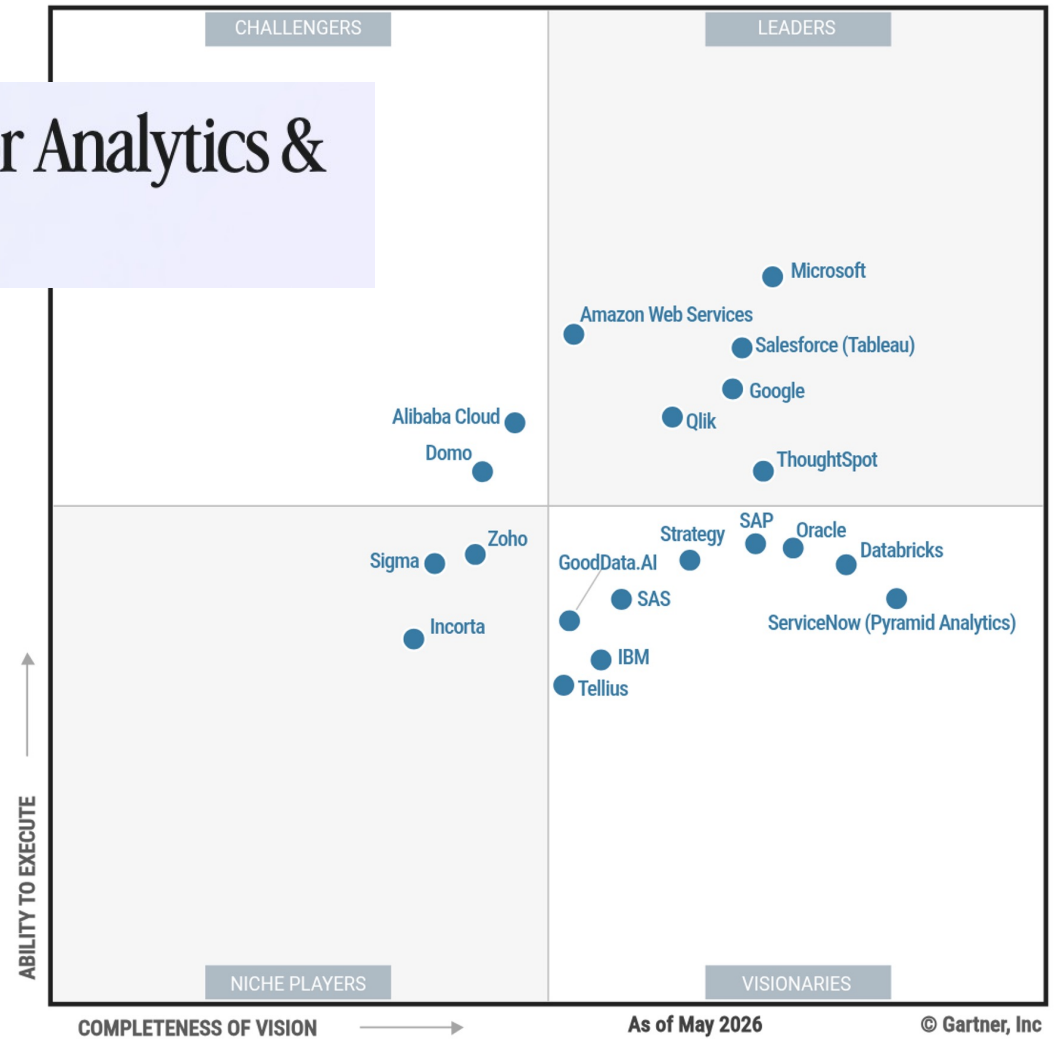
Power BI



tableau



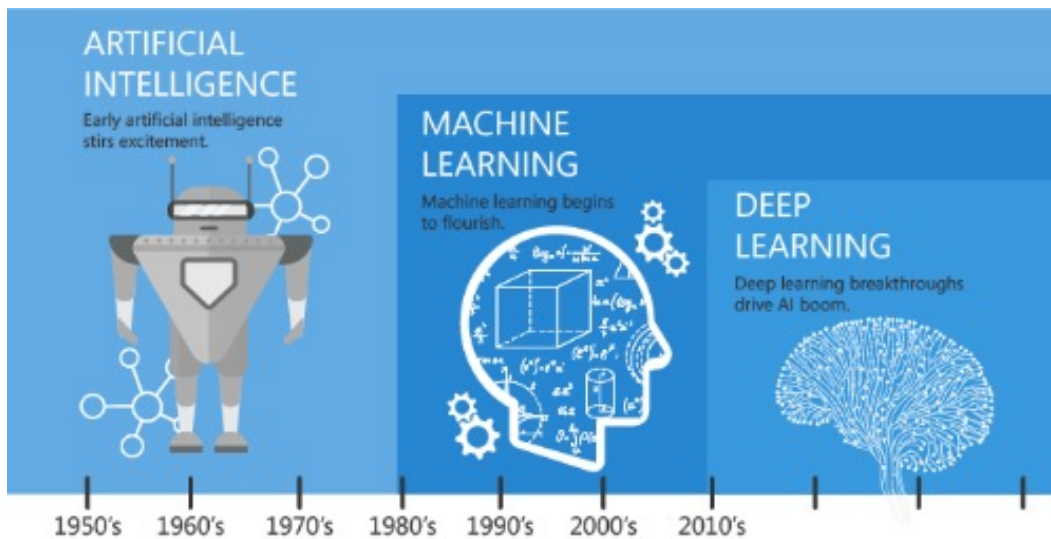
amazon
QuickSight



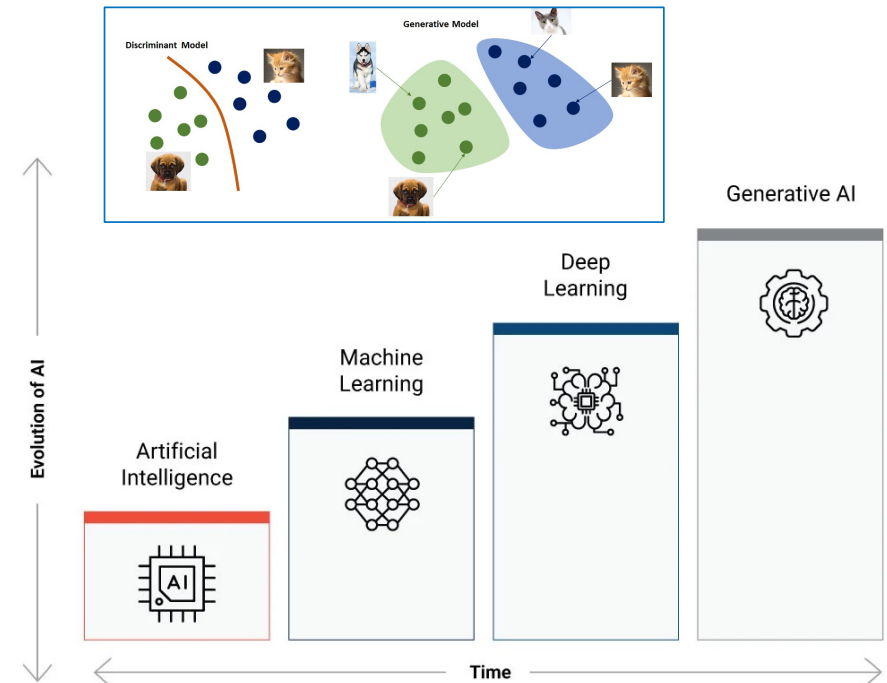
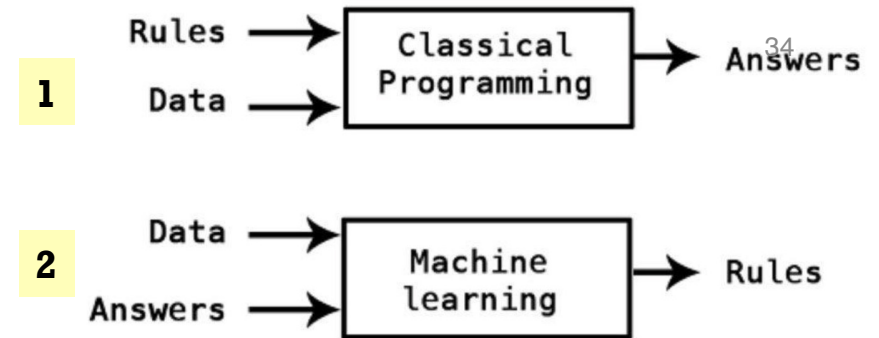


AI = Automation

- 1) Rule-based AI
- 2) Machine Learning (ML)



Since an early flush of optimism in the 1950's, smaller subsets of artificial intelligence - first machine learning, then deep learning, a subset of machine learning - have created ever larger disruptions.

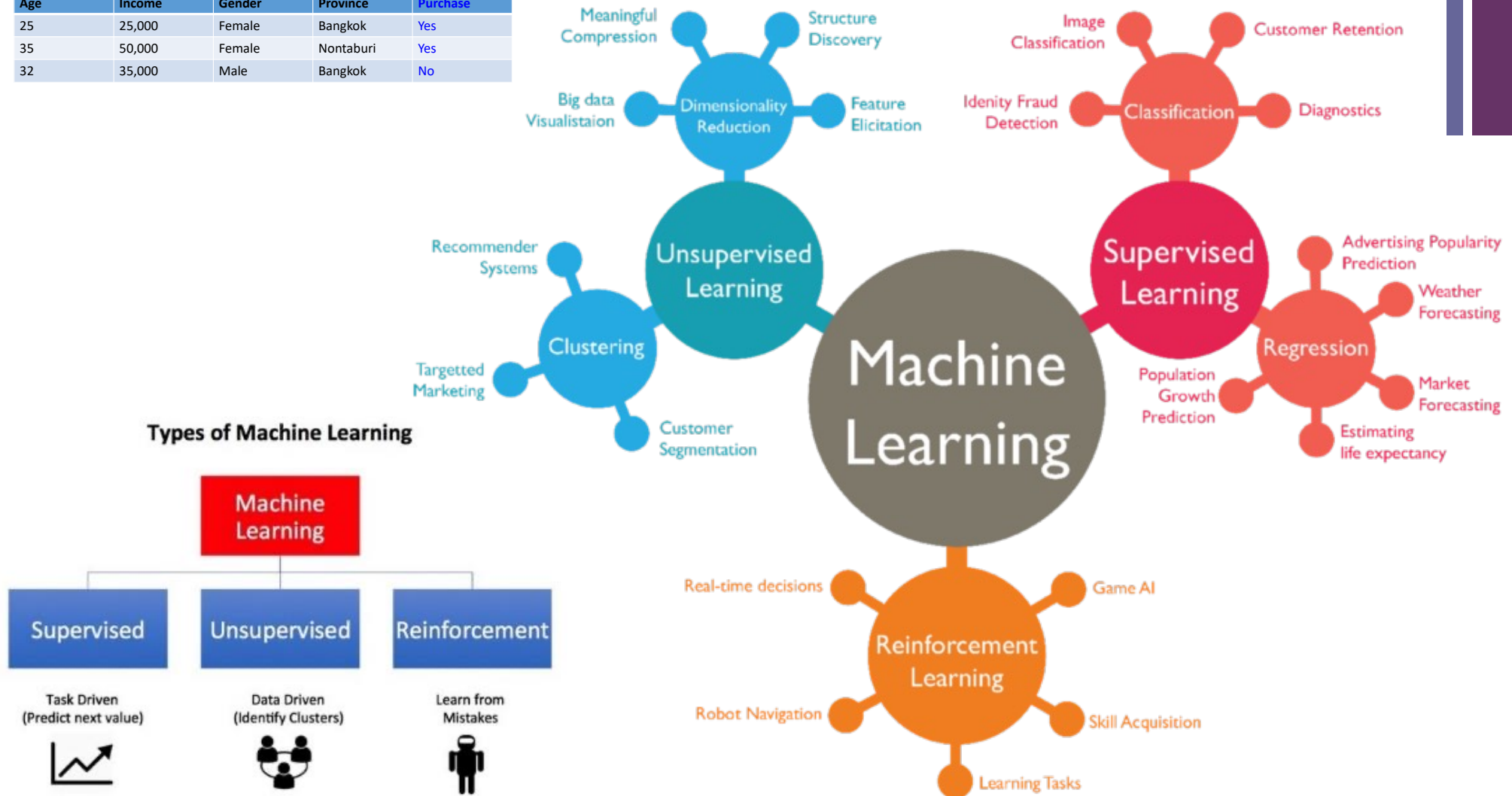


<https://mc.ai/machine-learning-basics-artificial-intelligence-machine-learning-and-deep-learning/>

+ Machine Learning (ML)

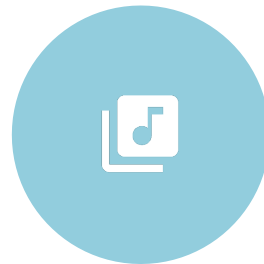
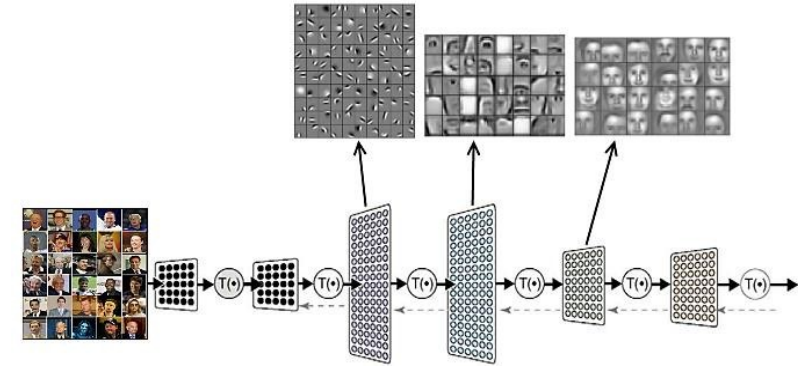
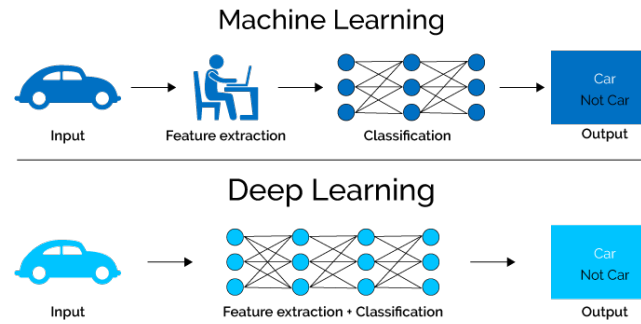
35

Age	Income	Gender	Province	Purchase
25	25,000	Female	Bangkok	Yes
35	50,000	Female	Nontaburi	Yes
32	35,000	Male	Bangkok	No





Deep Learning (DL)



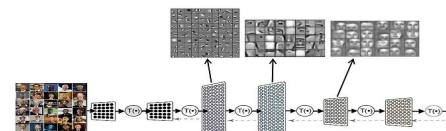
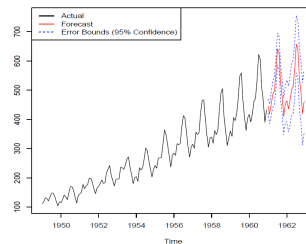
Speech
Recognition



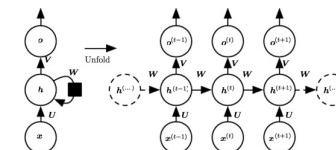
Computer
Vision



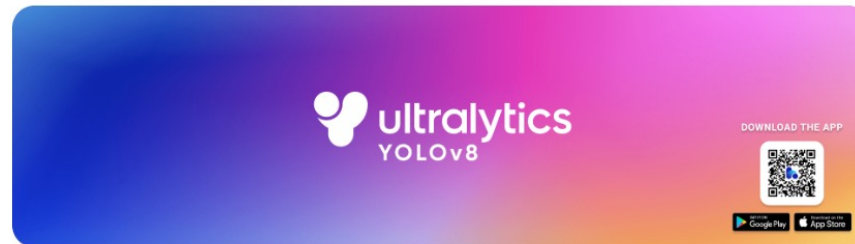
Natural Language
Processing



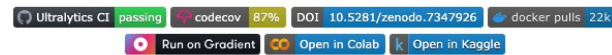
CNN



RNN (LSTM)



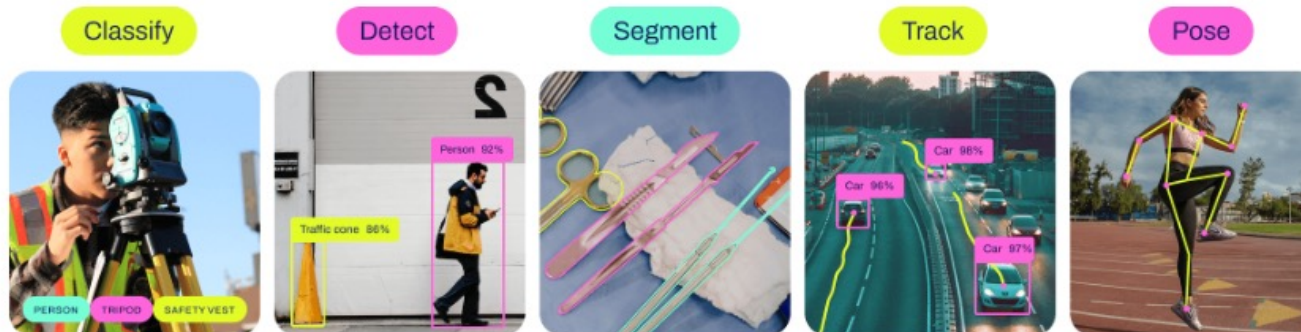
English | 简体中文



Ultralytics YOLOv8 is a cutting-edge, state-of-the-art (SOTA) model that builds upon the success of previous YOLO versions and introduces new features and improvements to further boost performance and flexibility. YOLOv8 is designed to be fast, accurate, and easy to use, making it an excellent choice for a wide range of object detection and tracking, instance segmentation, image classification and pose estimation tasks.

Models

YOLOv8 **Detect**, **Segment** and **Pose** models pretrained on the **COCO** dataset are available here, as well as YOLOv8 **Classify** models pretrained on the **ImageNet** dataset. **Track** mode is available for all Detect, Segment and Pose models.

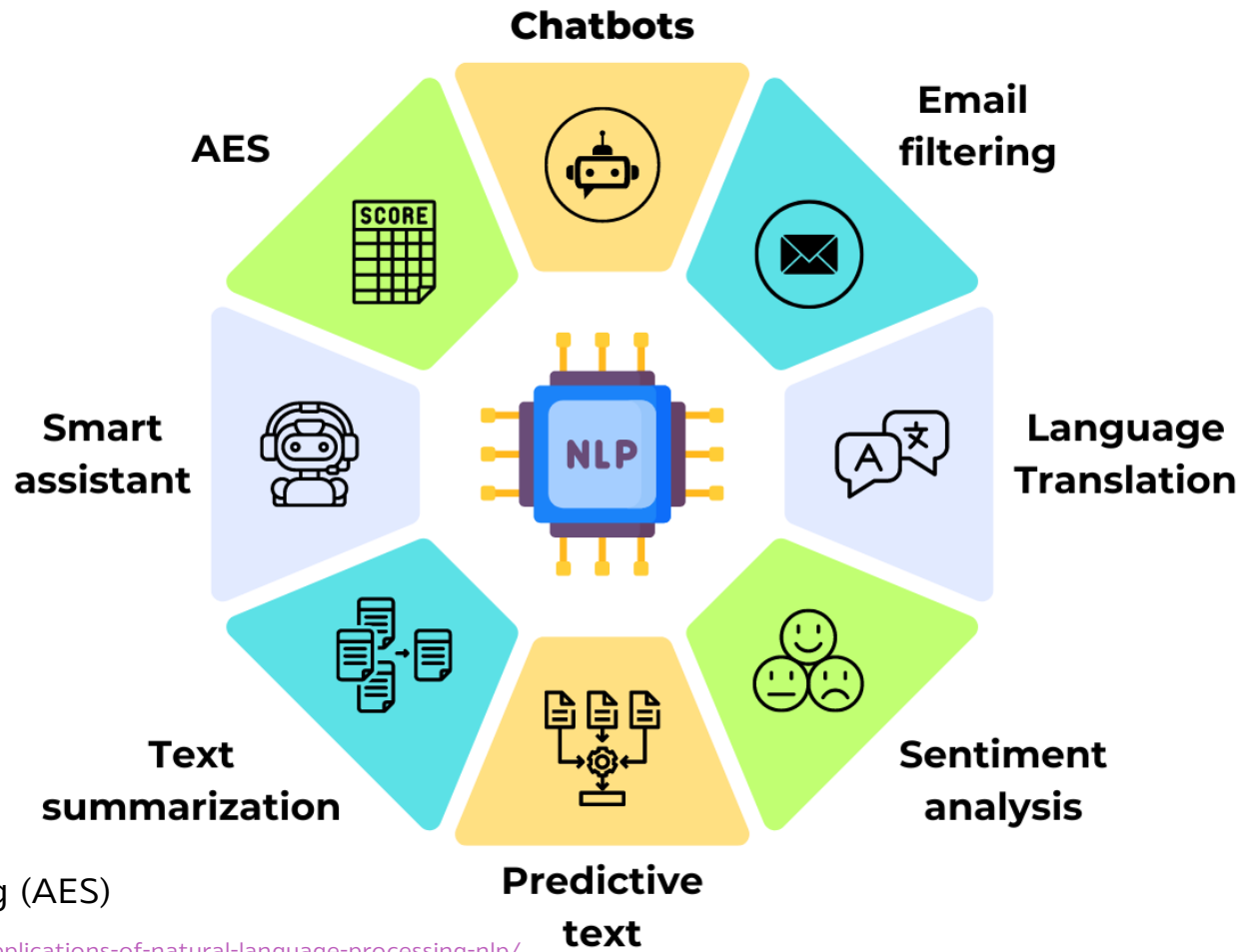


All **Models** download automatically from the latest Ultralytics **release** on first use.

<https://github.com/ultralytics/ultralytics>

Top 8 Applications of Natural Language Processing (NLP)

Applications of Natural Language Processing



<https://eastgate-software.com/top-8-applications-of-natural-language-processing-nlp/>



Automatic Speech Recognition (ASR) & Text-to-Speech (TTS)



Automatic Speech
Recognition



Natural Language
Processing



Text to
Speech



ChatGPT Surges in Popularity: Now Boasting Over 200 Million Weekly Users

Onome · September 2, 2024

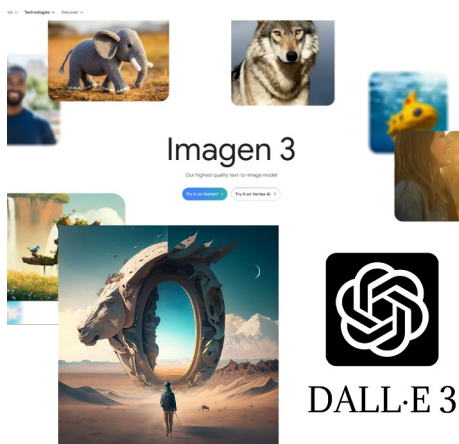
AI NEWS 3 MINS READ

<https://autogpt.net/chatgpt-surges-in-popularity-now-boasting-over-200-million-weekly-users/>



Generative AI

1) Image



DALL-E 3

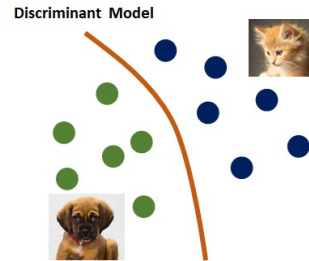
2) Text



Claude

Gemini

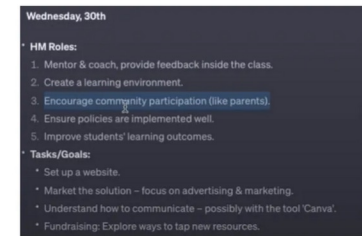
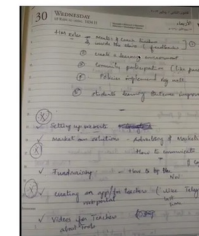
deepseek



3) Video



4) Multimodal



Prompt Engineering Techniques



Zero-shot
prompting



Few-shot prompting
or in-context
learning



Chain-of-thought
prompting



Tree-of-thought
prompting



Iterative
refinement



Feedback
loops



Prompt
chaining



Role-playing



Maieutic
prompting



Complexity-based
prompting

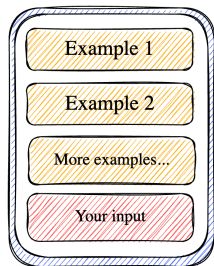
servicenow

<https://www.servicenow.com/ai/what-is-prompt-engineering.html>

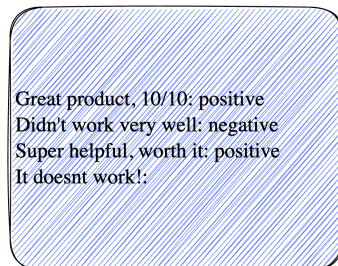


1) Few-Shot Prompting

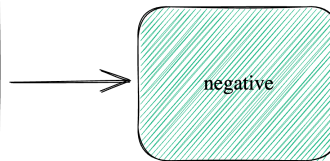
A Few Shot Prompt



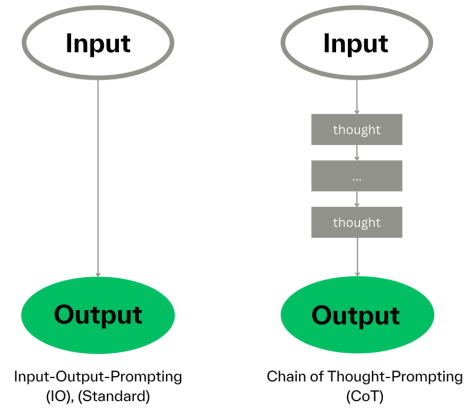
Example



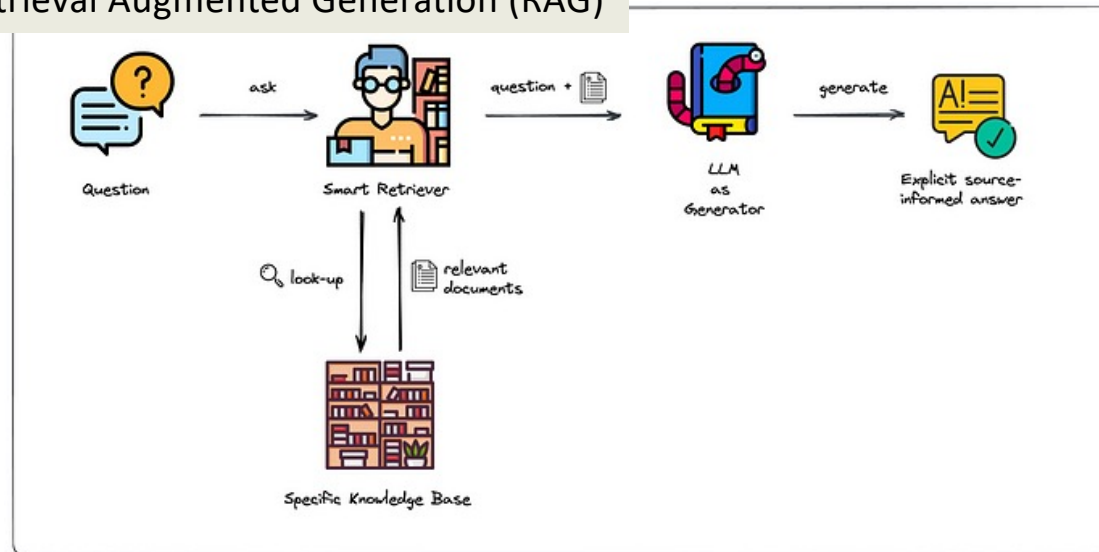
Model Output



2) Chain-of-Thoughts (CoT)



3) Retrieval Augmented Generation (RAG)



Jobs of the Future: AI Prompt Engineer



Cody W Burns

Emerging Technology Visionary | Distributed Systems | Privacy |
Executive Leadership

10 articles

+ Follow

October 19, 2022

**JOBS OF THE FUTURE:
AI PROMPT ENGINEER**

Cody Burns

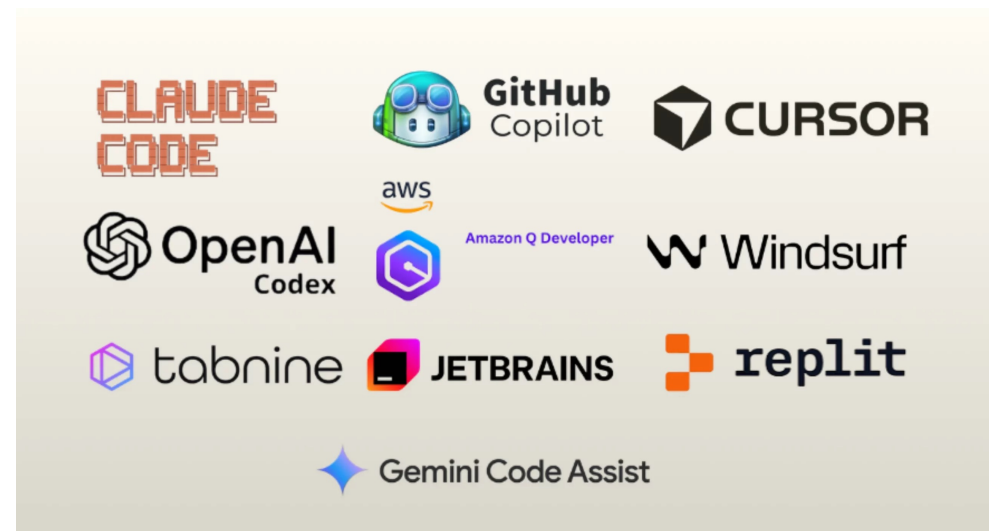
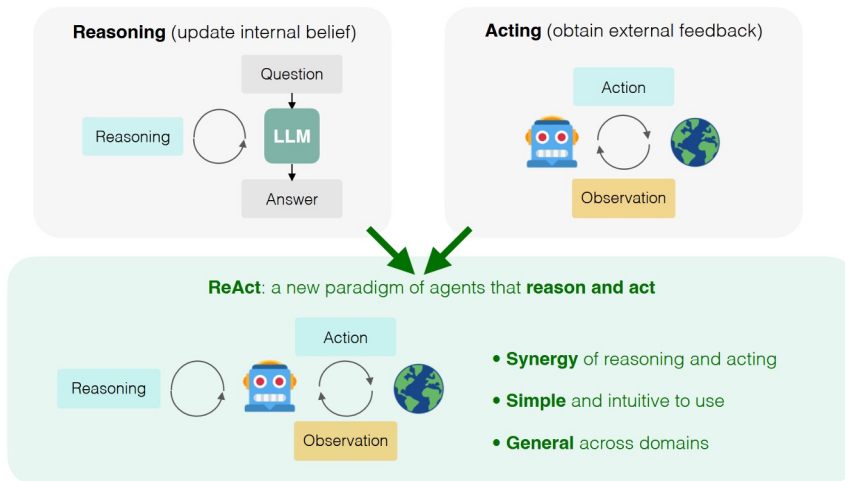
44

<https://www.linkedin.com/pulse/jobs-future-ai-prompt-engineer-cody-w-burns/>



Welcome to Agentic AI Era!

- AI is no longer just answering questions—it can understand instructions and take actions – **Agentic AI (think + code)**.
- For example, ChatGPT → **Codex = Coding + Execution**



+ Types of **Data** Science Projects (recap)

46

If we don't have data, can we still perform these tasks?

Valuable insights

- Data visualization
- Analytical skills & storytelling
 - Infographic



Advanced analytics

- AI/Machine Learning/Deep Learning
 - Prediction, Forecasting, Clustering, etc.





Data Engineering: Data Acquisition (e.g., SQL, web scraping) → Cleansing → Storing

47

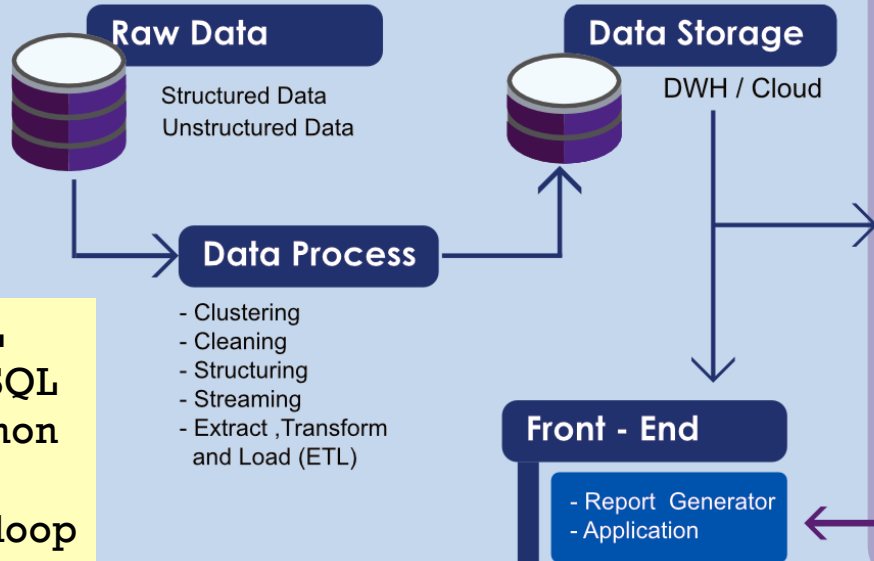
Steps of The Data Refinement Pipeline



Data Pipeline

invgate.com

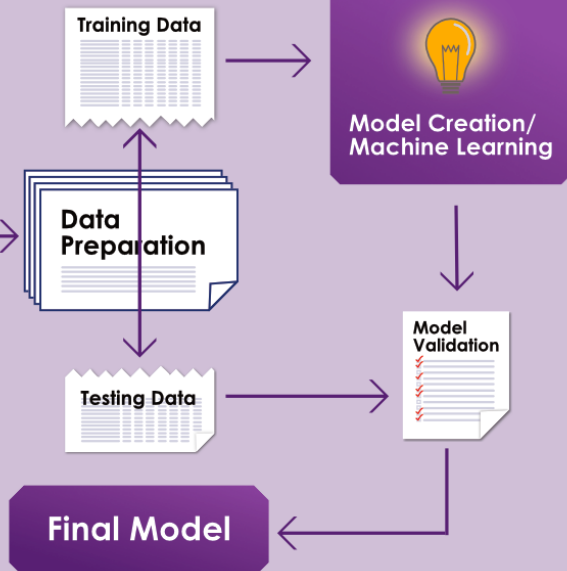
DATA ENGINEER



SQL
NoSQL
Python

Hadoop
Spark

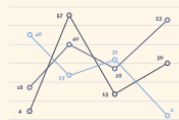
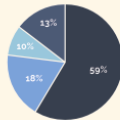
DATA SCIENTIST



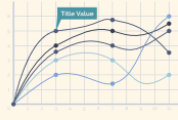
DATA ANALYST

Data Visualization
Bi
Application
Report
Action

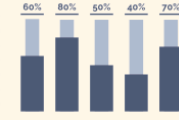
Nunc vitae eleifend massa. Quisque risi ex, viverra ut velit a, efficitur cursus nisi. Integer pretium risi turpis. Vitae sagittis nisi, metus luctus.



Your Title
Vivamus ultrices hendrerit neque eget tristique. Duis justo libero, hendrerit a turpis, nec, dictum pellentesque massa. Nunc vitae eleifend massa.



Your Title
Proin efficitur ornare luctus, vel pulvinar risus maximus nec. Lorem ipsum dolor sit amet.



Your Title
Proin efficitur ornare luctus, vel pulvinar risus maximus nec. Lorem ipsum dolor sit amet.



Your Title
Proin efficitur ornare luctus, vel pulvinar risus maximus nec. Lorem ipsum dolor sit amet.

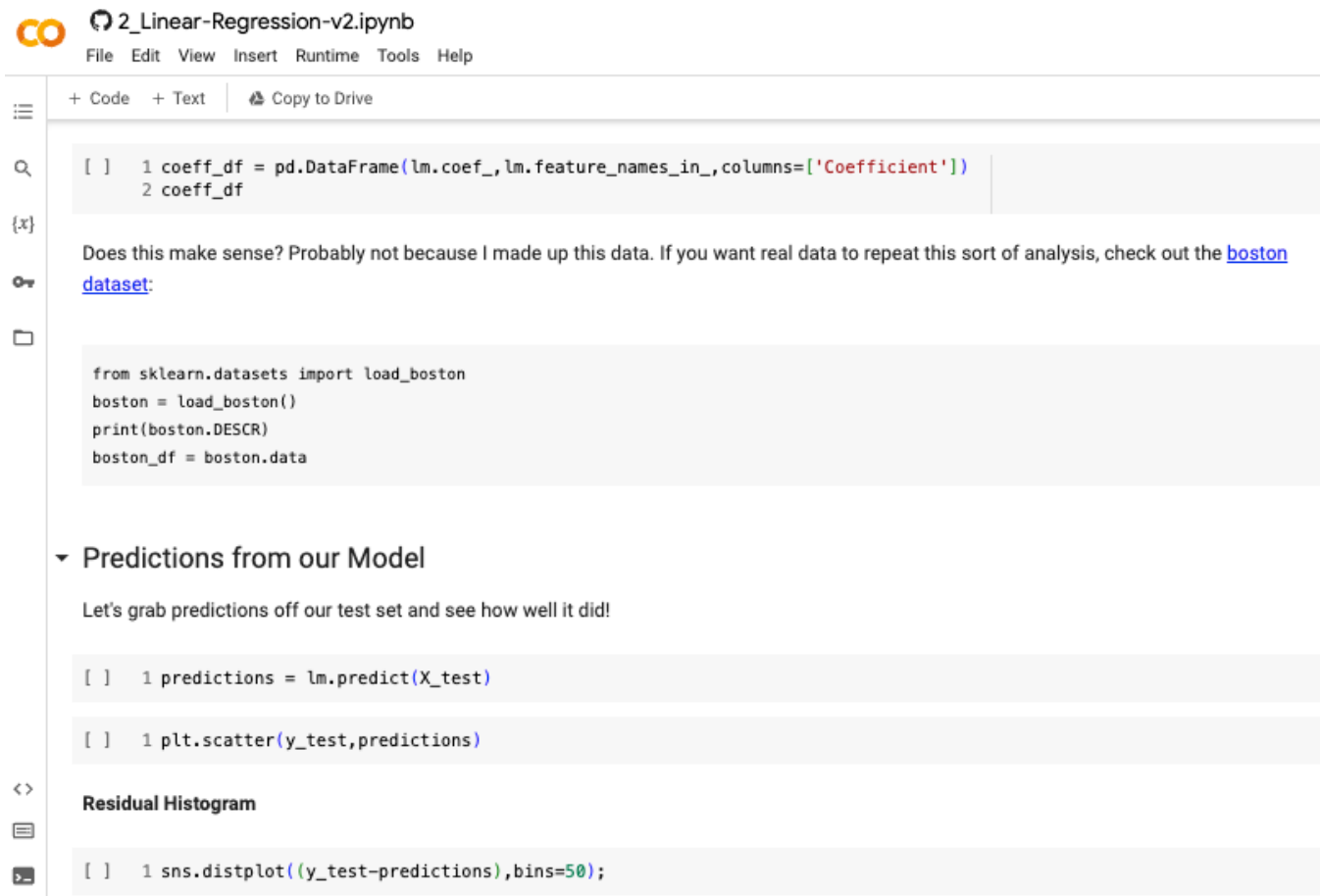
BigData Ecosystem



+ We have the data and the ML model.

So, are we all set? Or is there more to consider?

Do you think that user can use this code to get the prediction result? NO!!!



```
2_Linear-Regression-v2.ipynb
File Edit View Insert Runtime Tools Help

+ Code + Text Copy to Drive

[ ] 1 coeff_df = pd.DataFrame(lm.coef_, lm.feature_names_in_, columns=['Coefficient'])
    2 coeff_df

Does this make sense? Probably not because I made up this data. If you want real data to repeat this sort of analysis, check out the boston dataset:

from sklearn.datasets import load_boston
boston = load_boston()
print(boston.DESCR)
boston_df = boston.data

▼ Predictions from our Model

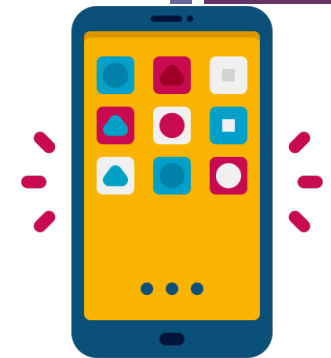
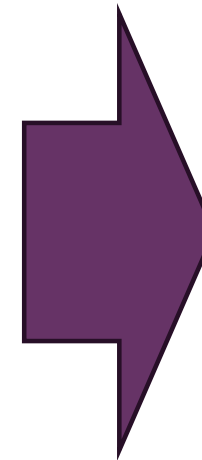
Let's grab predictions off our test set and see how well it did!

[ ] 1 predictions = lm.predict(X_test)

[ ] 1 plt.scatter(y_test, predictions)

Residual Histogram

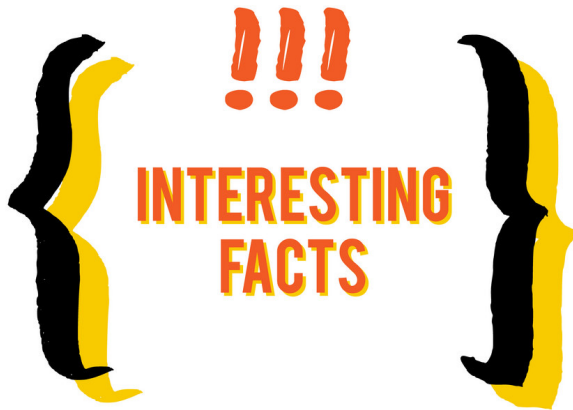
[ ] 1 sns.distplot((y_test-predictions), bins=50);
```





Interesting facts

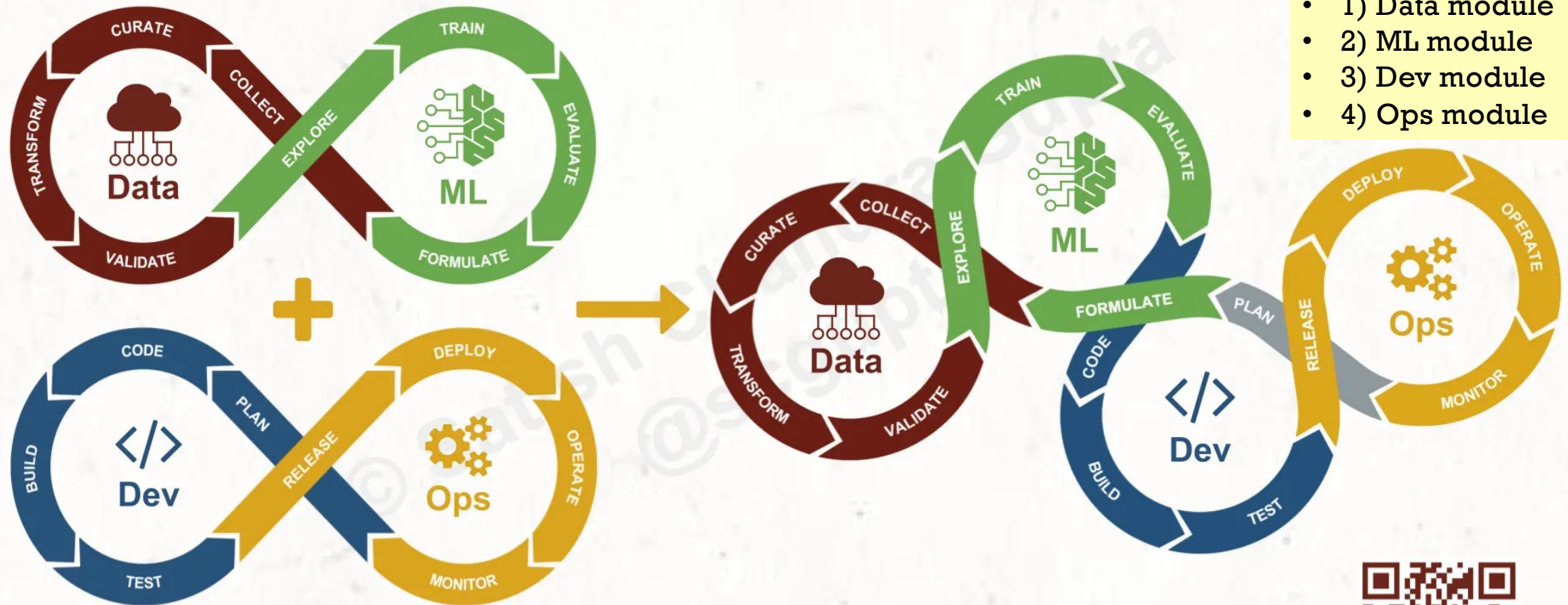
50



- More than 50% of AI projects were **FAILED** since they didn't plan about the deployment.
- | For the AI project, finish building the model is only 50% of the work.
- | The remaining work is about the deployment as a touchpoint to the target user.

MLOps = DataML + DevOps

ml4devs.com/mlops-lifecycle 



- 1) Data module
- 2) ML module
- 3) Dev module
- 4) Ops module

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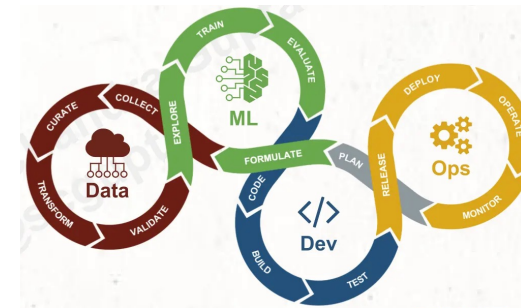
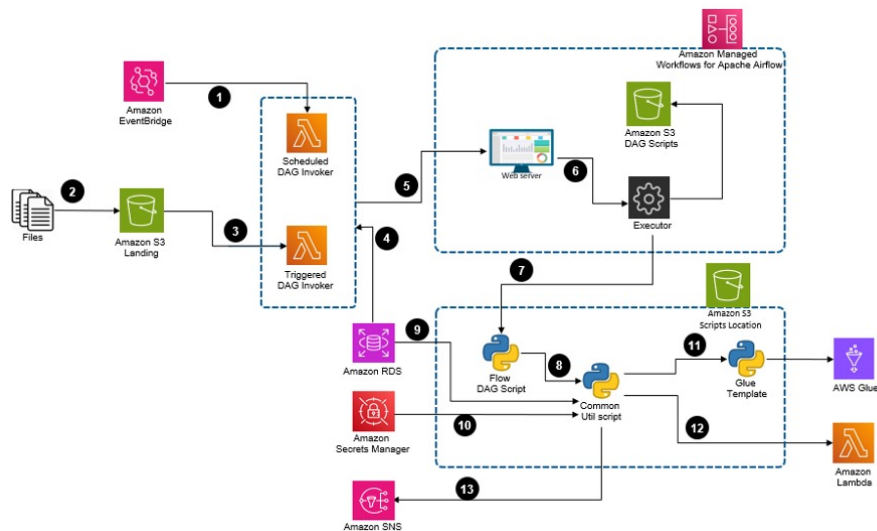
<https://www.ml4devs.com/images/illustrations/ml-lifecycle-fusing-model-and-software-development.webp>

scgupta.me 
twitter.com/scgupta 
linkedin.com/in/scgupta 

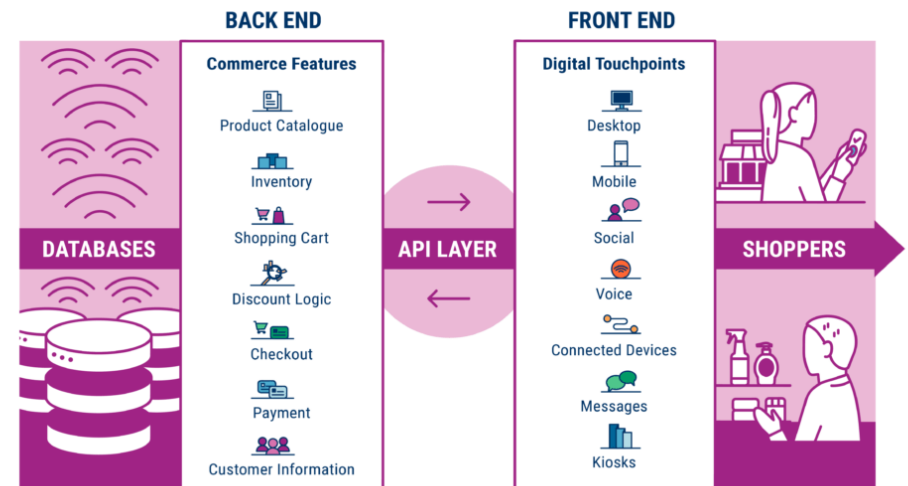
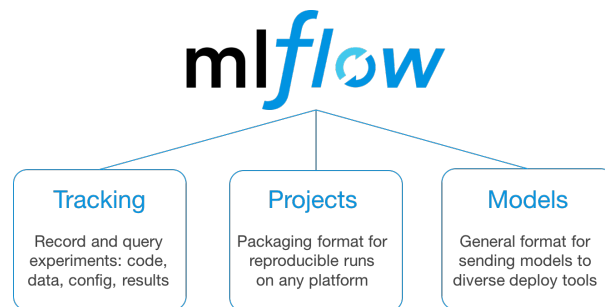


+ Many MLOps tasks: Deployment, Automation, Integration, Monitoring → Infrastructure + API

52



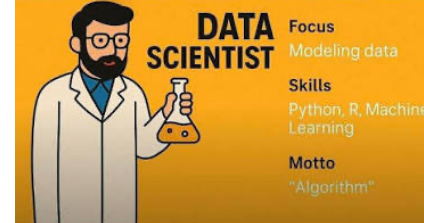
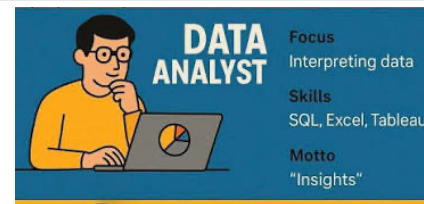
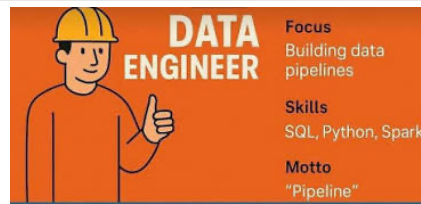
Components of a headless commerce architecture





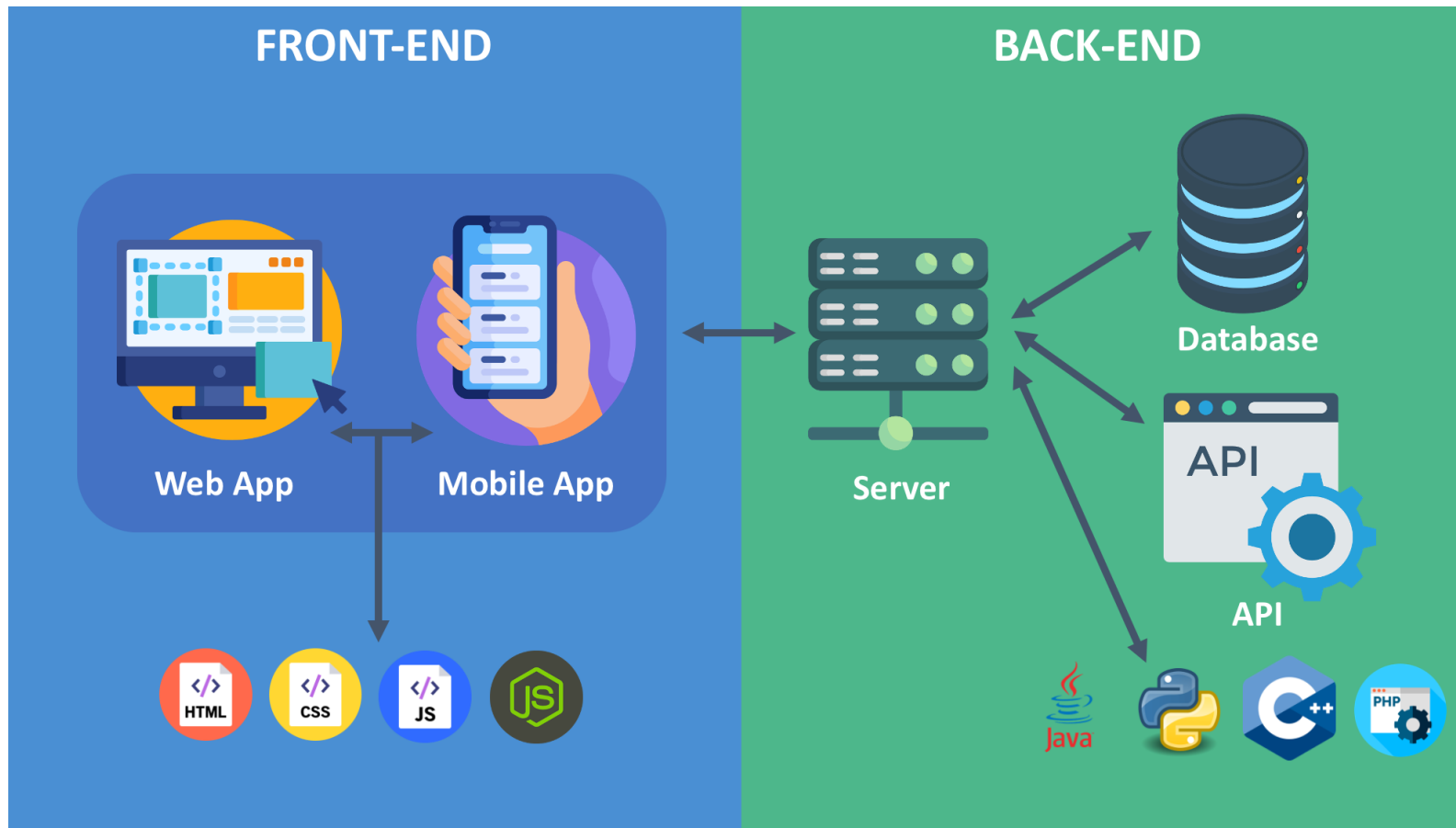
Data Engineer vs. Data Scientist vs. AI Engineer

Aspect	Data Engineer	Data Scientist	AI Engineer / ML Engineer
Focus	Data Infrastructure	Data Analytics	AI Systems
Goal	Prepare data Data Pipeline	Build models Model Pipeline	Deploy AI Model Deployment Pipeline
Key Tasks	ETL, Data Pipelines	ML, Statistics	DL, LLMs, MLOps
Output	Clean Data	Trained Models	AI Applications





Where should you develop and deploy your model or application? **Do you need to set up all the servers yourself?**





the 2026 Gartner® Magic Quadrant™ for Cloud AI Infrastructure



Figure 1: Magic Quadrant for Cloud AI Infrastructure



Categories of AWS services



Vit Niennattrakul, Ph.D.



Analytics



Application
Integration



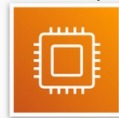
AR and VR



Blockchain



Business
Applications



Compute



Cost
Management



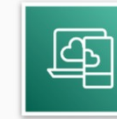
Customer
Engagement



Database



Developer Tools



End User
Computing



Game Tech



Internet
of Things



Machine
Learning



Management and
Governance



Media Services



Migration and
Transfer



Mobile



Networking and
Content Delivery



Robotics



Satellite



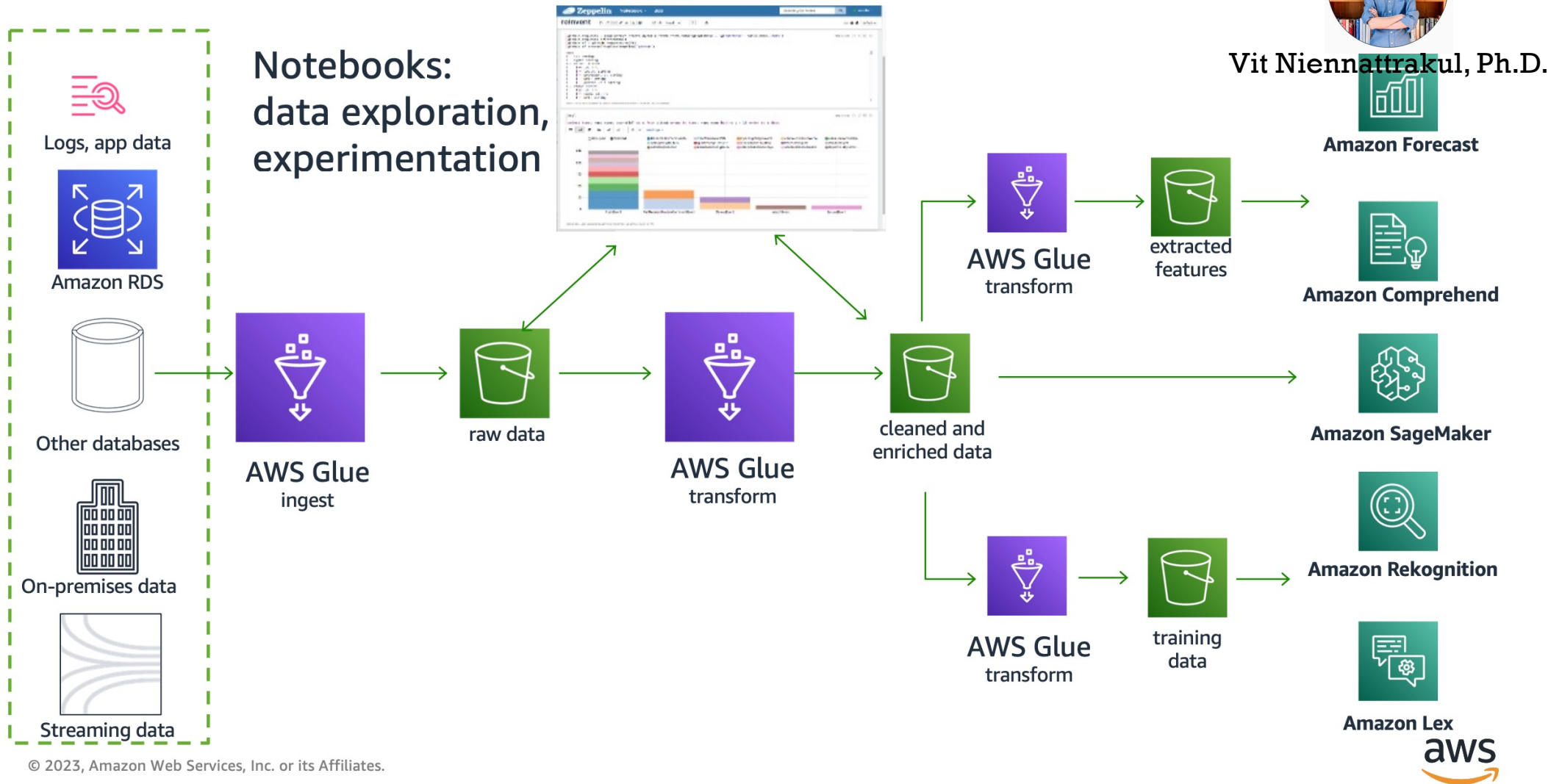
Security, Identity, and
Compliance



Storage

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Data preparation for AI / ML and data science



AWS Academy Service

AWS Academy Learner Lab

- Amazon API Gateway
- AWS App Mesh
- Application Auto Scaling
- AWS AppSync
- Amazon Athena
- Amazon Aurora
- AWS Backup
- AWS Certificate Manager (ACM)
- AWS Batch
- AWS Cloud9
- AWS CloudFormation
- Amazon CloudFront
- Amazon CloudSearch
- AWS CloudShell
- AWS CloudTrail
- Amazon CloudWatch
- AWS CodeCommit
- AWS CodeDeploy
- Amazon CodeWhisperer
- AWS Config
- AWS Systems Manager (SSM)
- Amazon Textract
- AWS Cost and Usage Report
- AWS Cost Explorer
- AWS Data Pipeline
- AWS DeepComposer
- AWS DeepLens
- AWS DeepRacer
- AWS Directory Service
- Amazon EC2 Auto Scaling
- AWS Elastic Beanstalk
- Amazon Elastic Block Store (EBS)
- Amazon Elastic Container Registry (ECR)
- Amazon Elastic Container Service (ECS)
- Amazon Elastic File System (EFS)
- Amazon Elastic Inference
- Amazon Elastic Kubernetes Service (EKS)
- Elastic Load Balancing (ELB)
- Amazon Elastic MapReduce (EMR)
- Amazon ElastiCache
- Amazon EventBridge
- AWS Fargate
- Amazon Timestream
- AWS Trusted Advisor
- Amazon Forecast
- AWS Glue
- AWS Glue DataBrew
- Amazon GuardDuty
- AWS Health
- AWS Identity and Access Management (IAM)
- AWS IAM Access Analyzer
- Amazon Inspector
- AWS IoT 1-Click
- AWS IoT Analytics
- AWS IoT Core
- AWS IoT Greengrass
- Amazon Kendra
- AWS Key Management Service (KMS)
- Amazon Kinesis
- Amazon Lex
- Amazon Machine Learning (Amazon ML)
- AWS Marketplace Subscriptions
- AWS Mobile Hub
- Amazon Neptune
- Amazon Virtual Private Cloud (Amazon VPC)
- AWS WAF - Web Application Firewall
- AWS OpsWorks
- Amazon Personalize
- Amazon QuickSight
- Amazon Redshift
- Amazon Relational Database Service (RDS)
- AWS Resource Groups & Tag Editor
- AWS RoboMaker
- Amazon Route 53
- AWS Secrets Manager
- AWS Security Hub
- AWS Security Token Service (STS)
- AWS Serverless Application Repository (SAR)
- AWS Service Catalog
- Amazon Simple Notification Service (SNS)
- Amazon Simple Queue Service (SQS)
- Amazon Simple Storage Service (S3)
- Amazon Simple Storage Service Glacier (S3 Glacier)
- Amazon Simple Workflow Service (SWF)
- AWS Step Functions
- AWS Storage Gateway
- AWS Well-Architected Tool
- AWS X-Ray

AWS Academy Lab Project - Cloud Data Pipeline Builder

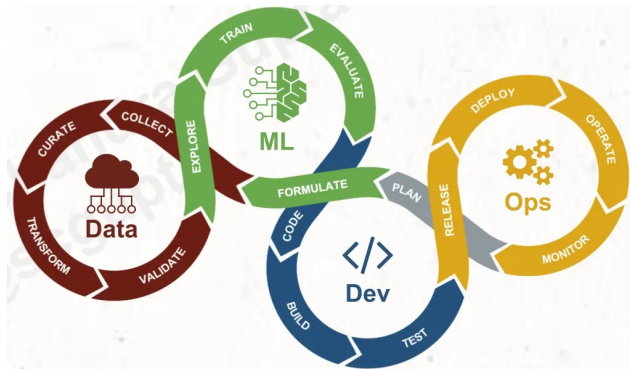
- Amazon Managed Streaming for Apache Kafka (Amazon MSK)

Both Learner Lab & Lab Project - Cloud Data Pipeline Builder

- Amazon SageMaker
- Amazon Elastic Compute Cloud (EC2)
- Amazon DynamoDB
- AWS Lambda
- Amazon Kinesis Video Streams
- Amazon Rekognition



Conclusion

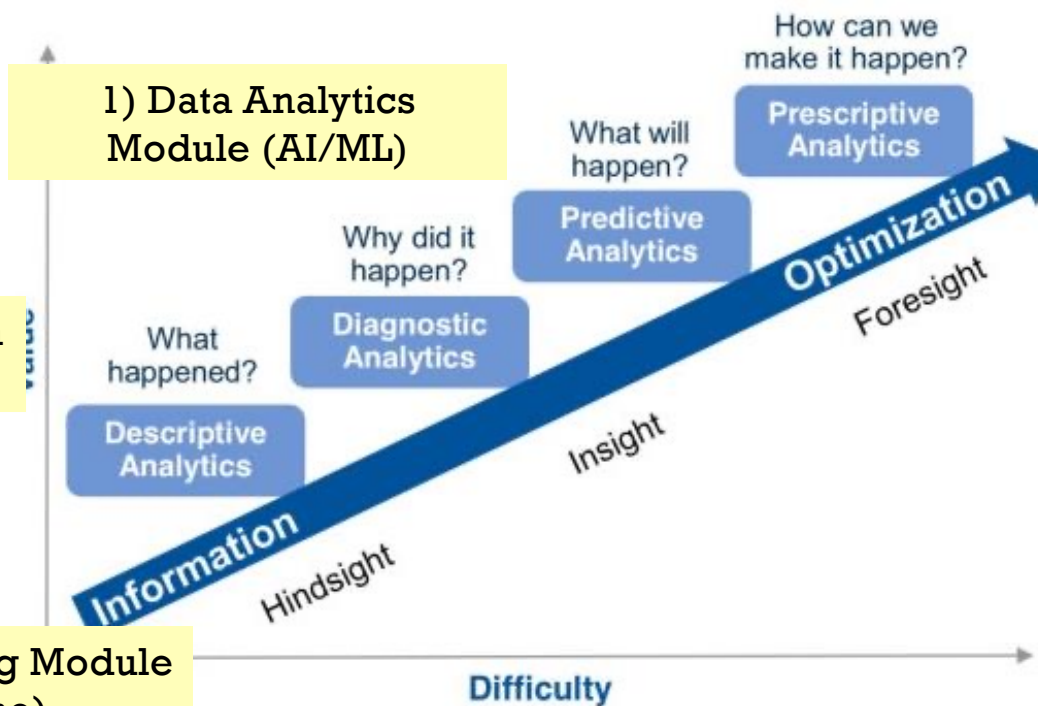


Conclusion

1) Data Analytics Module (AI/ML)

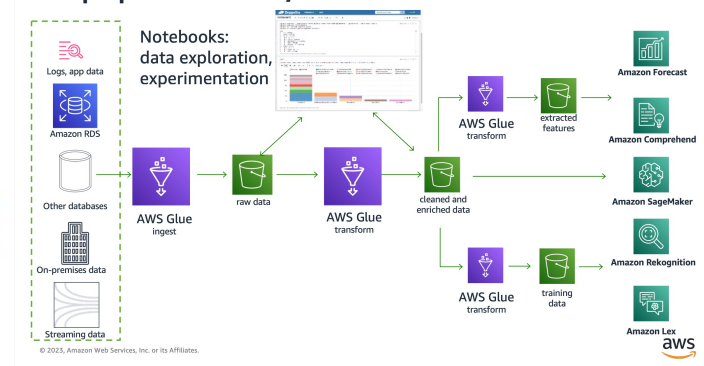
3) Data Vizualiation Module

2) Data Engineering Module (Data Pipeline)



4) Cloud Technology

Data preparation for AI / ML and data science





Any questions? 😊